Janaks Theorem and Orbital Energy Functionals (in progress)

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Abstract

Janaks theorem shows that the rate of change of the total N-electron energy with respect to the occupation numbers of the natural orbitals of the model Kohn-Sham (KS) independent particle state are equal to the Lagrange multipliers associated with the normalization constraint imposed on these orbitals. However many conceptual questions surround this theorem. In what sense can the occupation numbers vary as the model KS state is by definition an Independent Particle State (IPS) with integer occupation numbers? What is the functional relationship between the eigenvalues and the occupation number? How does one define a KS functional that is differentiable with respect to occupation numbers? In order to examine the meaning and content of this theorem we review the construction of the implicitly defined Hohenberg-Kohn (HK) and KS energy functionals and obtain an expression of Janaks theorem in the context of constrained optimization theory. It is shown that, in fact, Janaks theorem holds for any total energy functional defined in terms of orbitals and occupation numbers. In particular it applies to Antisymmetrized Geminal Power energy functionals, which also solely depend on orbitals and occupation numbers and whose functional form is explicitly known. The properties and structure of these orbital functionals are contrasted and compared and subtle differences are discussed..

Contents

I. INTRODUCTION

Janaks theorem [1] is a useful auxiliary tool in the Kuhn-Sham (KS) [2] approach to Density Functional Theory (DFT), that is helpful in analyzing the structure and meaning of the eigenvalues of the KS Fock operator. It states that these eigenvalues are equal to the derivatives of the exact total energy with respect to the occupation numbers of the First Order Reduced Density Operator (FORDO) of the model KS state. However many conceptual questions surround this theorem.

- In what sense can the occupation numbers vary as the model KS state is by definition an Independent Particle State (IPS) with integer occupation numbers?
- What is the functional relationship between the eigenvalues and the occupation number?
- How does one define a KS functional that is differentiable with respect to occupation numbers?

In both KS-DFT and Hartree-Fock (HF) theory these eigenvalues correspond to Lagrange parameters associated with the normalization constraint on the orbitals, that constitute the

Independent Particle State (IPS), and are related to the ionization energies of electrons described by these orbitals in the HF case and to the highest occupied one in the KS case. This correspondence can be generalized to Multi Configurational Self Consistent States (MCSCF) as described, for instance, in [3].

Constrained optimization theory [4] shows, if the Hessian of the Lagrangian is nonsingular then the values of Lagrange parameters correspond to the sensitivity of the object function with respect to the control parameters, i.e. to the values of the constraints. In this case the object function is the N-electron energy and the control parameters (that vary) are the norms of the orbitals. Hence Janaks theorem implies that *the orbitals must be normalized to their occupation numbers* and the energy is dependent on the occupation numbers through its dependence on these orbitals. It is of course possible to express the energy functional in terms of orthonormal orbitals and freely varying occupation numbers, that comply to the basic constraints of being positive with a maximum value of one, but this formulation does not directly introduce the relationship between the variation of the energy functional with respect to the occupation numbers and the eigenvalues of the Fock operator.

In general orbital functionals can be classified according to whether they are

- invariant to local phase changes of orbitals, i.e. are gauge invariant, thus preserve the electron density
- invariant to global phase changes of the orbitals i.e. to the group $U(1)$ [?], e.g. Density Matrix and KS Functionals
- not invariant to $U(1)$ e.g. correlated AGP functionals [3].

In order to answer the preceding questions and analyze the form of the energy dependence on occupation numbers all three types of orbital functionals will be discussed.

In this paper we will review some of the basic principles and underlying mathematical structure of Hohenburg-Kohn (HK) [5] and KS DFT using the formalism of super operators [6] (section II). This will allow us to express the Levy-Lieb construction of energy functionals as an eigenvalue problem. In section III we discuss orbital based functionals and in particular we consider KSDFT and derive Janaks Theorem in the context of a constrained variational approach and sensitivity theory again using the super operator formalism. The KS density functional, density matrix functional and the AGP functional determine the total energy in terms of a set of occupations numbers and one-electron orbitals. The first two have only been defined implicitly while the AGP functional has been explicitly constructed [3] in terms of General Spin Orbitals (GSOs), which we review in section IV. In this section we compare and contrast the properties of the derivatives of this functional with respect to occupation numbers to the derivatives obtained indirectly through sensitivity theory for the orbital based functional introduced in section III. In section V we conclude with a discussion and summary.

II. HOHENBERG-KOHN DFT

In this section we review some fundamental background needed to define the HK density functional utilizing the formalism of *super operators* [6] applied to a constrained optimization procedure that allows one to formulate its construction in terms of the solution of an eigenvalue problem. Most of the important technical mathematical aspects that are needed are examined in the appendices, but in some cases a more more detailed analysis is deferred elsewhere.

The HK density functional is defined in terms of the kinetic energy operator, K, the coulomb operator, V_2, and the charge density ρ, we first consider the properties of the charge density:

A. Densities and States

Density Functional Theory is based on the existence of a universal functional, $\mathcal{F}(\rho)$, of the charge density, $\rho(\boldsymbol{r})$, where

$$\rho(\boldsymbol{r}) = \int D^1(\boldsymbol{r}, \boldsymbol{\xi}; \boldsymbol{r}, \boldsymbol{\xi})\, d\boldsymbol{\xi} \tag{1}$$

and $D^1(\boldsymbol{r}, \boldsymbol{\xi}; \boldsymbol{r}', \boldsymbol{\xi}')$ is the integral kernel of the First Order Reduced Operator (FORDO) D^1 (see Appendix A for details of density and reduced density operators and matrices), which can be expanded as

$$D^1 = \int D^1(\boldsymbol{r}, \boldsymbol{\xi}; \boldsymbol{r}', \boldsymbol{\xi}')\, |\boldsymbol{r}, \boldsymbol{\xi}\rangle \langle \boldsymbol{r}', \boldsymbol{\xi}'|\, d\boldsymbol{r} d\boldsymbol{r}' d\boldsymbol{\xi} d\boldsymbol{\xi}' \tag{2}$$

in the basis $\{|\boldsymbol{r}, \boldsymbol{\xi}\rangle\}$ formed by the tensor product of the spin states $\{|\boldsymbol{\xi}\rangle\}$ and the generalized eigenvectors, $\{|\boldsymbol{r}\rangle\}$ (see Appendix B), of the position operator. In the proceeding we will use the notation ρ_D for a charge density that has been determined by a fixed state D and ρ for fixed charge density that could be associated with a set of states.

For a system that contains on average N-electrons the position density must satisfy the conditions

$$\int \rho\left(\boldsymbol{r}\right) d\boldsymbol{r} = N \Rightarrow \rho \in L_1\left(\mathbb{R}^3\right) \tag{3a}$$

$$\rho\left(\boldsymbol{r}\right) \geq 0 \tag{3b}$$

$$\rho \in L_3\left(\mathbb{R}^3\right). \tag{3c}$$

The function spaces $L_1\left(\mathbb{R}^3\right)$ and $L_3\left(\mathbb{R}^3\right)$ are defined in [7]. The constraint eq. $(3a)$ does not restrict the density to be the contraction of a N-electron state and one could consider incoherent and coherent mixtures of states describing different number of electrons. However in this article we primarily focus on N-electron states and mainly consider convex sets, $\left\{[\rho]_N\right\}$, of N-electron state density operators defined by

$$[\rho]_N = \left\{ D \middle| \Xi_1\left(D\right) = \rho; D \in S_N \subset \mathcal{B}_1\left(\mathcal{H}^N\right)\right\}, \quad 1 \leq N < \infty \tag{4}$$

(see Appendix A for the definition of the convex set S_N and of the Banach space of trace class operators $\mathcal{B}_1\left(\mathcal{H}^N\right)$ and eq. (6) for the definition of Ξ_1). The constraint eq. $(3c)$ ensures that the expectation value of the kinetic energy operator is finite [7] and the constraint eq. $(3b)$ is a necessary and sufficient condition for representability i.e. that there exists an electronic state that contracts to this density. The constraints eqs. $(3a, 3c$ and $3b)$ define a convex set $\mathcal{D}$ of densities

$$\rho \in \mathfrak{D} \subset L_{3+}\left(\mathbb{R}^3\right) \cap L_{1+}\left(\mathbb{R}^3\right), \tag{5}$$

(the "+" subscript denoting positive elements) that can be contracted from states that have a finite kinetic energy. The constraining functional Ξ_1 can be written in terms of the set of density operators $\left\{ \Phi\left(\boldsymbol{r}\right)^\dagger \Phi\left(\boldsymbol{r}\right) \middle| \boldsymbol{r} \in \mathbb{R}^3 \right\}$, (see Appendix C for their definition) as

$$\left[\Xi_1\left(D\right)\right]\left(\boldsymbol{r}\right) = \mathrm{Tr}\left\{\Phi\left(\boldsymbol{r}\right)^\dagger \Phi\left(\boldsymbol{r}\right) D\right\} = \rho_D\left(\boldsymbol{r}\right), \tag{6}$$

the [] notation emphasizes the roles of the different arguments D and $\boldsymbol{r}$, displaying that the map

$$\Xi_1 : \mathcal{B}_1\left(\mathcal{H}^N\right) \to L_1\left(\mathbb{R}^3\right) \tag{7}$$

has the value $[\Xi_1(D)] \in L_1(\mathbb{R}^3)$ at the point D and the value $[\Xi_1(D)](\boldsymbol{r}) = \mathrm{Tr}\left\{\Phi(\boldsymbol{r})^\dagger \Phi(\boldsymbol{r}) D\right\} \in \mathbb{R}$ at the physical position $\boldsymbol{r}$. In order to examine the properties of Ξ_1 it is convenient and more useful to consider the range of Ξ_1 to be contained in the space $L_1(\mathbb{R}^3)$ rather than in $L_3(\mathbb{R}^3)$ or $\mathfrak{D}$ itself. The value, $\Xi_1(D)$, of the function Ξ_1 at the point D can be expressed in terms of an operator valued distribution, $\Phi^\dagger\Phi_\delta$, whose domain is restricted to the set of Dirac delta functions, $\mathcal{D}_\delta$, defined on $\mathbb{R}^3$ with values in the space, $\mathcal{B}\left(\mathcal{H}^N\right)$, of bounded operator (see Appendices A and C) i.e.

$$\Phi^\dagger\Phi_\delta : \mathcal{D}_\delta \to \mathcal{B}\left(\mathcal{H}^N\right) \tag{8}$$

$$\Phi^\dagger\Phi_\delta(\delta_{\mathbf{r}}) = \Phi(\boldsymbol{r})^\dagger \Phi(\boldsymbol{r}) \tag{9}$$

leading to

$$\Xi_1(D) = \mathrm{Tr}\left\{\Phi^\dagger\Phi_\delta D\right\} = \rho_D \in L_1\left(\mathbb{R}^3\right). \tag{10}$$

As D is a positive operator it can always be expressed as a square

$$D = A^2, \tag{11}$$

where

$$A \in \mathcal{B}_2\left(\mathcal{H}^N\right) \text{ and } A = A^\dagger \equiv A \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \tag{12}$$

and $\mathcal{B}_2\left(\mathcal{H}^N\right)$ and $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ are the Hilbert spaces of Hilbert-Schmidt and self adjoint Hilbert-Schmidt operators respectively acting on the N-electron space $\mathcal{H}^N$ (see appendix A for the definitions of $\mathcal{B}_2\left(\mathcal{H}^N\right)$ and $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$). The operator A is the square root of D, which is not unique, as can be observed from the spectral expansion

$$A = D^{\frac{1}{2}} = \sum_{1\le j\le m} \pm\sqrt{\alpha_j}\mathcal{P}_j, \tag{13}$$

where

$$D = \sum_{1\le j\le m} \alpha_j \mathcal{P}_j;\ \alpha_j \ge 0, 1 \le j \le m, \tag{14}$$

$r_N = \sum_{1\le j\le m} \nu_j =$ dimension of $\mathcal{H}^N$, $\{\nu_j | 1 \le j \le m\}$ the degeneracies of the eigenvalues $\{\alpha_j | 1 \le j \le m\}$ and $\{\mathcal{P}_j | 1 \le j \le m\}$ are orthogonal projectors onto the eigenspaces of D. If one restricts A to be positive, i.e. uses the positive square roots in eq. (13), then it belongs to the cone, $\mathcal{B}_{2sa+}\left(\mathcal{H}^N\right)$, [8] of positive Hilbert-Schmidt operators and uniquely determines D. However, in general it not convenient to restrict A to be positive and one allows the domain of variation of the N-electron energy functional to be contained in the *real* Hilbert space $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$, of dimension r_N^2, which has an orthonormal bases

$$\begin{aligned}
\Lambda_{jk} &= \frac{1}{\sqrt{2}}\left\{|\Psi_j\rangle\langle\Psi_k| + |\Psi_k\rangle\langle\Psi_j|\right\}; \quad 1 \le j < k \le r_N \\
\Lambda_{jj} &= |\Psi_j\rangle\langle\Psi_j|; \quad 1 \le j \le r_N \\
\Lambda_{jk} &= \frac{i}{\sqrt{2}}\left\{|\Psi_j\rangle\langle\Psi_k| - |\Psi_k\rangle\langle\Psi_j|\right\}; \quad 1 \le j > k \le r_N,
\end{aligned} \tag{15}$$

where $\{|\Psi_j\rangle| 1 \le j \le r_N\}$ is any complete orthonormal basis for $\mathcal{H}^N$, of course all these spaces are infinite dimensional if $r_N = \infty$.

B. The Density Constraint

The Levy-Leib construction [7, 9] is based on defining the value of the density functional as the minimum of an energy variation over states constrained to have a fixed density $\rho \in \mathfrak{D}$. The density constraint can be formulated in terms of a quadratic functional Ξ_2, based on the linear functional Ξ_1 eq. (6), acting on the space of self adjoint Hilbert-Schmidt operators, $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$, where

$$\begin{gathered}
\Xi_2 : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to \mathfrak{D} \subset L_{3+}\left(\mathbb{R}^3\right) \cap L_{1+}\left(\mathbb{R}^3\right) \subset L_1\left(\mathbb{R}^3\right) \\
\Xi_2(A) = \Xi_1\left(A^2\right) = \operatorname{Tr}\left\{A\Phi^\dagger\Phi_\delta A\right\} = \left(A\left|\Phi^\dagger\Phi_\delta A\right.\right) = \rho,
\end{gathered} \tag{16}$$

where the real symmetric inner product $(\cdot|\cdot)$ in $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ is defined as

$$(X|Y) = \operatorname{Tr}\{XY\}. \tag{17}$$

The constraint eq. (16) can be formulated in terms of the *super operator valued distribution* $\widehat{\Phi^\dagger\Phi_\delta}$, as

$$\Xi_2(A) = \left(A\left|\widehat{\Phi^\dagger\Phi_\delta}A\right.\right) = \rho, \tag{18}$$

where

$$\widehat{\Phi^\dagger\Phi_\delta} : \mathcal{D}_\delta \to \mathcal{B}\left(\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)\right), \tag{19}$$

and has components defined as (Appendix C)

$$\widehat{\Phi^\dagger\Phi_\delta}(\boldsymbol{r})|X) = \Phi\widehat{(\boldsymbol{r})^\dagger\Phi}(\boldsymbol{r})|X) = \left|\left[\Phi(\boldsymbol{r})^\dagger\Phi(\boldsymbol{r})X\right]_+\right) \ \forall X \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right), \boldsymbol{r} \in \mathbb{R}^3, \tag{20}$$

where

$$\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)X\right]_{+}=\frac{1}{2}\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right),X\right]_{+}. \tag{21}$$

The positive commutator $[,]_{+}$ is used in the definition of the super operator in order to ensure that its action produces a self adjoint operator.

The functional $\Xi_2(A)$ is Fréchet differentiable and one can form the differential, $d\Xi_2(A)$, [8], (Appendix E 1)

$$d\Xi_2(A):\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)\rightarrow L_1\left(\mathbb{R}^3\right) \tag{22}$$

of Ξ_2 at any point $A\in\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$, in terms of the components

$$\left[d\Xi_2(A)\right](\boldsymbol{r})=\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right|. \tag{23}$$

The set of operators $\left\{\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)\right|\boldsymbol{r}\in\mathbb{R}^3\right\}$ forms a generalized basis (Appendix B) for the normal spaces of the surface $\Xi_2(A)=\rho$ for all points A (Appendix E), which shows that $d\Xi_2(A)$ is surjective [8] and thus the constraint $\Xi_2(A)=\rho$ is regular [4] everywhere. *****

C. Hohenberg-Kohn Density Functional Construction

Even though the explicit form of the Hohenberg-Kohn functional, $\mathcal{F}_{HK}(\rho)$, is not known one can define it implicitly [7, 9] by

$$\mathcal{F}_{HK}(\rho)=\mathrm{Tr}\left\{(K+V_2)A_*(\rho)^2\right\}=E_U\left(A_*(\rho)\right) \tag{24}$$

in terms of the minima, $E_U\left(A_*(\rho)\right)$, of the universal energy functional $E_U(A)$, that are solutions of the constrained quadratic optimization problem

$$E_U\left(A_*(\rho)\right)=\min_{A\in\mathcal{B}_{2sa}(\mathcal{H}^N)}\mathrm{Tr}\left\{A(K+V_2)A\right\}=\left(A_*(\rho)\left|\left(\widehat{K}+\widehat{V}_2\right)A_*(\rho)\right.\right) \tag{25a}$$

$$\text{subject to }\Xi_2(A)=\left(A\left|\widehat{\Phi^{\dagger}\Phi}A\right.\right)=\rho, \tag{25b}$$

where K, and V_2 are the kinetic energy and coulomb operators, the corresponding super operators $\widehat{K}$ and $\widehat{V}_2$ are defined as

$$\widehat{S}\left|X\right)=\left|[SX]_{+}\right)\ \forall X\in\mathcal{B}_{2sa}\left(\mathcal{H}^N\right),\quad S=K,V_2 \tag{26}$$

and the control densities $\rho \in \mathcal{D}$. This choice of the density ρ coupled with insisting that A is an N-electron operator, i.e. $A \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$, ensures that A is normalized (Appendix G), i.e.

$$(A\,|A) = 1. \tag{27}$$

As eq. (25) is a convex problem a local minimum is a global minimum, which, however, may be degenerate. The Lagrangian associated with this problem eq. (25) is

$$\mathcal{L}\left(A, \lambda, \rho\right) = E_U\left(A\right) - \left\langle \lambda \right| \left\{\Xi_2\left(A\right) - \rho\right\}\rangle, \tag{28}$$

where the universal energy is

$$E_U\left(A\right) = \mathrm{Tr}\left\{\left(K + V_2\right) A^2\right\} = \left(A \middle| \left(\widehat{K} + \widehat{V_2}\right) A\right). \tag{29}$$

The Lagrange multiplier term can be expanded as

$$\left\langle \lambda \right| \left\{\Xi_2\left(A\right) - \rho\right\}\rangle = \int \lambda\left(\boldsymbol{r}\right)\left\{\Xi_2\left(A\right)\left(\boldsymbol{r}\right) - \rho\left(\boldsymbol{r}\right)\right\} d\boldsymbol{r},\ \lambda \in L_\infty\left(\mathbb{R}^3\right) \oplus L_{\frac{3}{2}}\left(\mathbb{R}^3\right), \rho \in \mathcal{D}, \tag{30}$$

where $\langle \cdot | \cdot \rangle$ is the scalar product between the dual spaces $\left\langle L_\infty\left(\mathbb{R}^3\right) \oplus L_{\frac{3}{2}}\left(\mathbb{R}^3\right), L_1\left(\mathbb{R}^3\right) \cap L_3\left(\mathbb{R}^3\right)\right\rangle$ i.e.

$$\langle \cdot | \cdot \rangle : \left(L_\infty\left(\mathbb{R}^3\right) \oplus L_{\frac{3}{2}}\left(\mathbb{R}^3\right)\right) \times \left(L_1\left(\mathbb{R}^3\right) \cap L_3\left(\mathbb{R}^3\right)\right) \to \mathbb{R}. \tag{31}$$

This constraint term in the Lagrangian eq. (28) can also be expressed in the form

$$\left\langle \lambda \right| \left\{\Xi_2\left(A\right) - \rho\right\}\rangle = \langle \lambda\,|\rho\rangle - \left(A \middle| \widehat{\Omega\left(\lambda\right)}\, A\right) = \left(A \middle| \left\{\langle \lambda\,|\rho\rangle\, \hat{I} - \widehat{\Omega\left(\lambda\right)}\right\} A\right), \tag{32}$$

where $\hat{I}$ is the super operator identity operator, $\Omega\left(\lambda\right)$ is a local one body potential defined as

$$\Omega\left(\lambda\right) = \int \lambda\left(\boldsymbol{r}\right) \boldsymbol{\Phi}\left(\boldsymbol{r}\right)^\dagger \boldsymbol{\Phi}\left(\boldsymbol{r}\right) d\boldsymbol{r} \tag{33}$$

and the associated super operator expectation values are given by

$$\left(A \middle| \widehat{\Omega\left(\lambda\right)}\, A\right) = \int \lambda\left(\boldsymbol{r}\right) \Xi_2\left(A\right)\left(\boldsymbol{r}\right) d\boldsymbol{r}. \tag{34}$$

The Lagrangian for a fixed N-particle density ρ can then be recast as

$$\mathcal{L}\left(A, \lambda, \rho\right) = \left(A \middle| \left(\widehat{K} + \widehat{V_2} - \widehat{\Omega\left(\lambda\right)}\right) A\right) + \langle \lambda\,|\rho\rangle. \tag{35}$$

D. Optimization Conditions

Solutions , A_*, of the constrained quadratic optimization eq. (25) correspond to unconstrained local saddle points (A_*, λ_*) [4] of the Lagrangian $\mathcal{L}(A, \lambda, \rho)$

$$\mathcal{L}(A_*, \lambda_*, \rho) = \max_{\lambda} \min_{A} \mathcal{L}(A, \lambda, \rho) \tag{36}$$

that satisfy the necessary condition

$$d\mathcal{L}(A_*, \lambda_*, \rho) = 0, \tag{37}$$

where the linear map for a fixed ρ

$$d\mathcal{L}(A, \lambda, \rho) : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \times \left(L_\infty\left(\mathbb{R}^3\right) \oplus L_{\frac{3}{2}}\left(\mathbb{R}^3\right)\right) \to \mathbb{R} \tag{38}$$

is given by

$$d\mathcal{L}(A, \lambda, \rho)(h, \varphi) = \langle \varphi \,|\rho\rangle - \left(A \left| \widehat{\Omega(\varphi)}\; A\right.\right) + 2\left(h \left| \left\{\widehat{K} + \widehat{V}_2 - \widehat{\Omega(\lambda)}\right\} A\right.\right). \tag{39}$$

As the constraint is regular the Lagrange functional λ_* is unique [4], but there may be degenerate A_*'s. The differential eq. (39) can be partitioned into $(d_A\mathcal{L}(A, \lambda, \rho), d_\lambda\mathcal{L}(A, \lambda, \rho))$, where

$$d_A\mathcal{L}(A, \lambda, \rho)(\cdot) = 2\left(\cdot \left| \left\{\widehat{K} + \widehat{V}_2 - \widehat{\Omega(\lambda)}\right\} A\right.\right) \tag{40a}$$

$$d_\lambda\mathcal{L}(A, \lambda, \rho)(\cdot) = \langle \cdot \,|\rho\rangle - (A \,|\Omega(\cdot)\; A), \tag{40b}$$

thus giving the stationary point conditions

$$\left(h \left| \left\{\widehat{K} + \widehat{V}_2 - \widehat{\Omega(\lambda_*)}\right\} A_*\right.\right) = 0\, \forall h \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \tag{41a}$$

$$\langle \varphi \,|\rho\rangle - \left(A_* \left| \widehat{\Omega(\varphi)}\; A_*\right.\right) = 0\, \forall \varphi \in L_\infty\left(\mathbb{R}^3\right) \oplus L_{\frac{3}{2}}\left(\mathbb{R}^3\right). \tag{41b}$$

If eq. $(41a)$ and eq. $(41b)$ are true for all operators h belonging to $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ and all functions φ belonging to $L_\infty(\mathbb{R}^3) \oplus L_{\frac{3}{2}}(\mathbb{R}^3)$ these conditions are equivalent to

$$\left\{\widehat{K} + \widehat{V}_2 - \widehat{\Omega(\lambda_*)}\right\} |A_*) = 0 \tag{42}$$

and the constraining condition

$$\left(A_* \left| \widehat{\Phi^\dagger \Phi}\; A_*\right.\right) = \rho. \tag{43}$$

(For unbounded operators $K + V_2 - \Omega(\lambda_*)$ one must be more precise, but here we will only consider the case when an eigenvector $|A_*)$ exists and thus eq. $(41a)$ is true for all $h \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ and ρ is V-representable, more general cases will be discussed elsewhere.

⌈

$$\left\{\widehat{K} + \widehat{V}_2 - \widehat{\Omega(\lambda_*)}\right\} |A_*) = 0 \Rightarrow \mathrm{Tr}\left\{h\left\{K + V_2 - \Omega(\lambda_*)\right\} A_*\right\} = 0 \forall h$$

⌋

The optimal value of the Lagrangian satisfies the following identities:

$$\begin{aligned}\mathcal{L}(A_*, \lambda_*, \rho) &= \left(A_*(\rho)\middle|\left\{\left\{\widehat{K} + \widehat{V}_2 - \widehat{\Omega(\lambda_*)}\right\} + \langle\lambda_*(\rho)|\,\rho\rangle\,\hat{I}\right\} A_*(\rho)\right) = \langle\lambda_*(\rho)|\,\rho\rangle \\ &= \left(A_*(\rho)\middle|\left\{\widehat{K} + \widehat{V}_2\right\} A_*(\rho)\right) = E_U(A_*(\rho)). \end{aligned} \tag{44}$$

This shows that the optimal values of the universal energy functional, the value of the Lagrangian and the value of the HK energy functional are all equal at the density ρ i.e.

$$\mathcal{F}_{HK}(\rho) = E_U(A_*(\rho)) = \langle\lambda_*(\rho)|\,\rho\rangle. \tag{45}$$

Moreover ρ is the density of the ground state $D_*(\rho) = A_*(\rho)^2$ of the Hamiltonian

$$K + V_2 - \Omega(\lambda_*(\rho)) \tag{46}$$

i.e. of an electronic system subject to an external potential $\lambda_*(\rho)$. Note that the ground state might be a ensemble state.

E. Sensitivity Analysis

Sensitivity analysis [4] examines how the optimal solution changes with variation of the value of the constraining functions, which can be thought of as control parameters. In this case the N-electron state energy with variations of ρ. If the optimal points $\{A_*, \lambda_*, \rho\}$ are regular (Appendix E) for a set of $\rho' s$ then $\{A_*, \lambda_*\}$ are in fact functions of ρ, i.e.

$$\begin{aligned} A_* &= A_*(\rho) \\ \lambda_* &= \lambda_*(\rho), \end{aligned} \tag{47}$$

and thus single valued i.e. the ground state is non degenerate. The sensitivity of the optimal N-electron energy to changes in the density ρ is then given by

$$\frac{\delta E_U(A_*(\rho))}{\delta\rho} = \lambda_*(\rho). \tag{48}$$

By construction

$$\mathcal{F}_{HK}(\rho) = E_U(A_*(\rho)), \tag{49}$$

which shows

$$\frac{\delta\mathcal{F}_{HK}(\rho)}{\delta\rho} = \lambda_*(\rho) \tag{50}$$

i.e. that the external potential is equal to the functional derivative of the HK functional with respect to the density.

F. Decomposition of the Hohenburg-Kohn Functional

If a minimizer $D_*(\rho)$ exists and is unique and then one can define kinetic energy, coulomb and exchange-correlation functionals as

$$\begin{aligned}\mathcal{F}_K(\rho) &= \mathrm{Tr}\{KD_*(\rho)\}, \\ \mathcal{F}_C(\rho) &= \int \frac{\rho(\boldsymbol{r},\boldsymbol{\xi})\,\rho(\boldsymbol{r}',\boldsymbol{\xi})}{\|\boldsymbol{r}-\boldsymbol{r}'\|} d\boldsymbol{r}d\boldsymbol{r}'d\boldsymbol{\xi}, \\ \mathcal{F}_{XC}(\rho) &= \mathrm{Tr}\{V_2D_*(\rho)\} - \mathcal{F}_C(\rho),\end{aligned} \tag{51}$$

producing the decomposition

$$\mathcal{F}_{HK}(\rho) = \mathcal{F}_K(\rho) + \mathcal{F}_C(\rho) + \mathcal{F}_{XC}(\rho). \tag{52}$$

The concept of a unique exchange-correlation functional is central to the theory, but the existence of non isolated degenerate $D_*(\rho)'$s could give different $\mathcal{F}_{XC}(\rho)'$s and $\mathcal{F}_K(\rho)'$s. However if degeneracies do occur one can still define an exchange-correlation functional through the imposition of further constrained minimizations.

G. System Energy

The total energy functional of a system that contains N electrons and experiences an environment described by a *local* potential v is

$$E(\rho,\mathrm{v}) = \mathcal{F}_{HK}(\rho) + f_\rho(\mathrm{v}), \tag{53}$$

where the explicit effect of the environment is described by the functional

$$f_\rho(\mathrm{v}) = \int \mathrm{v}(\boldsymbol{r})\,\rho(\boldsymbol{r})\,d\boldsymbol{r}. \tag{54}$$

The ground state energy $E_{GS}(\mathrm{v})$ and the corresponding density $\rho_{GS}(\mathrm{v})$ are then determined by the solution to the constrained optimization problem

$$E_{GS}(\mathrm{v}) = E(\rho_{GS},\mathrm{v}) = \min_{\rho\in\mathcal{D}} E(\rho,\mathrm{v}), \tag{55}$$

which one can make unconstrained in terms of real functions $f \in H_1(\mathbb{R}^3)$ [7] by setting

$$\rho(\boldsymbol{r}) = \frac{N f(\boldsymbol{r})^2}{\langle f|f\rangle} = \frac{N\left\langle f \left| \Phi(\boldsymbol{r})^\dagger \Phi(\boldsymbol{r}) f \right.\right\rangle}{\langle f|f\rangle},\ f \in H_1\left(\mathbb{R}^3\right). \tag{56}$$

The minimizer f_{GS} of $E(f,\mathrm{v})$ then satisfies

$$\frac{\delta E(f_{GS},\mathrm{v})}{\delta f(\boldsymbol{r})} = 2 f_{GS}(\boldsymbol{r}) \left\{ \frac{N}{\langle f_{GS}|f_{GS}\rangle} \frac{\delta E(f_{GS},\mathrm{v})}{\delta\rho(\boldsymbol{r})} - \int \frac{\delta E(f_{GS},\mathrm{v})}{\delta\rho(\boldsymbol{r}')} \rho_{GS}(\boldsymbol{r}')\, d\boldsymbol{r}' \right\} \overset{a.e.}{=} 0 \tag{57}$$

which implies that

$$f_{GS}(\boldsymbol{r}) \overset{a.e.}{=} 0 \Leftrightarrow \rho_{GS}(\boldsymbol{r}) \overset{a.e.}{=} 0 \tag{58}$$

or

$$\frac{\delta E(f_{GS},\mathrm{v})}{\delta\rho(\boldsymbol{r})} \overset{a.e.}{=} \alpha(\rho_{GS}), \tag{59}$$

where

$$\alpha(\rho_{GS}) = \frac{\langle f_{GS}|f_{GS}\rangle}{N} \int \frac{\delta E(f_{GS},\mathrm{v})}{\delta\rho(\boldsymbol{r}')} \rho_{GS}(\boldsymbol{r}')\, d\boldsymbol{r}'. \tag{60}$$

Hence

$$\alpha(\rho_{GS}) = \frac{\langle f_{GS}|f_{GS}\rangle}{N} \alpha(\rho_{GS}) \int \rho_{GS}(\boldsymbol{r}')\, d\boldsymbol{r}' = \frac{\langle f_{GS}|f_{GS}\rangle}{N} \alpha(\rho_{GS}) \tag{61}$$

$$\Rightarrow \alpha(\rho_{GS}) = 0 \text{ or } f_{GS}(\boldsymbol{r}) \overset{a.e.}{=} 0. \tag{62}$$

So as $f_{GS}(\boldsymbol{r}) \overset{a.e.}{\neq} 0$ we obtain the well known result

$$\frac{\delta E(f_{GS},\mathrm{v})}{\delta\rho(\boldsymbol{r})} \overset{a.e.}{=} 0 \Leftrightarrow \frac{\delta\mathcal{F}_{HK}(\rho_{GS})}{\delta\rho(\boldsymbol{r})} \overset{a.e.}{=} -\mathrm{v}(\boldsymbol{r}). \tag{63}$$

III. ORBITAL VARIABLES

There are two standard approaches to formulating the total energy in terms of orbitals and occupation numbers, one is to consider functionals of FORDO's, which leads to DMFT, the other is to factor the dependence through the density, which leads to KS-DFT and is based on the HK functional. However as the explicit forms of these functionals are not known one must rely on the implicit definitions via constrained optimizations, though in practice approximate explicit forms are utilized. We have recently [3] introduced a third approach which is defined on states generated by the submanifold of $2M$-electron AGP states. The total energy functional, which is explicitly given, *is more general* than a density matrix functional while still expressing the total energy in terms of orbitals and occupation numbers.

The central focus of this article is on functionals of the density that contain no explicit reference to the spin properties of the system. However one could widen this focus by considering spin and general spin densities that are described by vectors $\boldsymbol{\rho}(\boldsymbol{r}) = (\rho_\alpha(\boldsymbol{r}), \rho_\beta(\boldsymbol{r}))$ and $\boldsymbol{\rho}(\boldsymbol{r}) = (\rho_\alpha(\boldsymbol{r}), \rho_{\alpha\beta}(\boldsymbol{r}), \rho_\beta(\boldsymbol{r}))$ respectively and modify the preceding analysis in the appropriate fashion (Appendix I).

A. Correspondence between Orbitals and Densities

Any density ρ can expressed in terms of the Natural Orbitals (NO's) $\{\psi_j\}$ and occupation numbers $\{N_j\}$ of the FORDO $D_*^1(\rho)$ (Appendix A) as

$$\rho(\boldsymbol{r}) = \left[D_*^1(\rho)\right](\boldsymbol{r}, \boldsymbol{r}) = \sum_{1 \le j \le r} N_j \left|\psi_j(\boldsymbol{r})\right|^2, \tag{64}$$

but many other sets of orbitals also produce the same density, in particular Harimanns [10] theorem ensures that there are always expansions of the form

$$\rho(\boldsymbol{r}) = \left[D_*^1(\rho)\right](\boldsymbol{r}, \boldsymbol{r}) = \sum_{1 \le j \le N} \chi_j(\boldsymbol{r})^2 \tag{65}$$

in terms of real orthonormal square summable functions $\{\chi_j | 1 \le j \le N\}$.

In general one can express the density in terms of orthogonal square integrable functions as

$$\rho(\boldsymbol{r}) = \sum_{1 \le j \le \nu} \left|\varphi_j(\boldsymbol{r})\right|^2, \tag{66}$$

where

$$\langle \varphi_j | \varphi_k \rangle = N_j \delta_{jk}; 0 < N_j \le N, 1 \le j \le \nu; N_j = 0, \nu + 1 \le j \le r \tag{67}$$

and $\nu \equiv \nu(\boldsymbol{\varphi})$ is called the rank of the expansion. If $\nu \ge N$ then the functions $\boldsymbol{\varphi} = (\varphi_1, \cdots, \varphi_r)$ can be interpretted as orbitals. One can also expand the density ρ in terms of orthonormal functions $\tilde{\boldsymbol{\varphi}}$ and weights $\boldsymbol{N} = (N_1, \cdots, N_r)$ as

$$\rho(\boldsymbol{r}) = \sum_{1 \le j \le \nu} N_j \left|\tilde{\varphi}_j(\boldsymbol{r})\right|^2 . \tag{68}$$

Again if $\nu \ge N$ then $\tilde{\boldsymbol{\varphi}}$ and $\boldsymbol{N}$ can be interpretted as normalized orbitals and occupation numbers respectively. One can make the relationship expressed eq. (66) and eq. (68) explicit through the introduction of two functionals Γ_1 and Γ_2 :

- The first functional Γ_1 is defined by

  $$\Gamma_1 : \mathfrak{S}_1 \subset \prod_{1 \le j \le r} \mathcal{H}^1 \to L_{1+}\left(\mathbb{R}^3\right) \subset L_1\left(\mathbb{R}^3\right), \tag{69}$$

  where

  $$\rho = \Gamma_1(\boldsymbol{\varphi}), \tag{70a}$$
  $$\left[\Gamma_1(\boldsymbol{\varphi})\right](\boldsymbol{r}) = \sum_{1 \le j \le r} \left|\varphi_j(\boldsymbol{r})\right|^2 = \rho(\boldsymbol{r}) \tag{70b}$$

  and

  $$\mathfrak{S}_1 = \left\{\boldsymbol{\varphi} \left| \varphi_j, \varphi_k \in \mathcal{H}^1, \langle \varphi_j | \varphi_k \rangle = N_j \delta_{jk}, \boldsymbol{N} \in \mathfrak{N}(N) \right.\right\}, \tag{71}$$

  where

  $$\mathfrak{N}(N) = \left\{ \boldsymbol{N} \left| 0 \le N_j \le N, 1 \le j \le r; \sum_{1 \le j \le r} N_j = N \right.\right\}. \tag{72}$$

  and as many as $r-1$ of the component functions, φ_j, may be identically zero. The set $\mathfrak{S}_1$ includes all possible expansions of ρ even the one term case considered in eq. (56).

- The second functional Γ_2 is defined by

  $$\Gamma_2 : \mathfrak{S}_2 \subset \prod_{1 \le j \le r} \mathcal{H}^1 \times \mathbb{R}^r \to L_{1+}\left(\mathbb{R}^3\right) \subset L_1\left(\mathbb{R}^3\right), \tag{73}$$

  where

  $$\mathfrak{S}_2 = \left\{(\tilde{\boldsymbol{\varphi}}, \boldsymbol{N}) \left| |\tilde{\varphi}_j\rangle, |\tilde{\varphi}_k\rangle \in \mathcal{H}^1, \langle \varphi_j | \varphi_k \rangle = \delta_{jk}, 1 \le j \le r; \quad \boldsymbol{N} \in \mathfrak{N}(N) \right.\right\}. \tag{74}$$

  Clearly it is always possible to find $\boldsymbol{\varphi}$ and $(\tilde{\boldsymbol{\varphi}}, \boldsymbol{N})$ such that

  $$\rho = \Gamma_1(\boldsymbol{\varphi}) = \Gamma_2(\tilde{\boldsymbol{\varphi}}, \boldsymbol{N}) \tag{75}$$

and if the number of non zero components in $\boldsymbol{\varphi}$ and $\boldsymbol{N}$ are greater than or equal to N then $\boldsymbol{\varphi}$ and $\{\tilde{\boldsymbol{\varphi}}, \boldsymbol{N}\}$ determine the same FORDO D^1 i.e.

$$D^1(\boldsymbol{\varphi}) = D^1(\tilde{\boldsymbol{\varphi}}, \boldsymbol{N}) = \sum_{1\leq j\leq r} |\varphi_j\rangle\langle\varphi_j| = \sum_{1\leq j\leq r} |\tilde{\varphi}_j\rangle\langle\tilde{\varphi}_j| N_j. \tag{76}$$

The optimization wrt ρ can thus be expressed in terms of $\boldsymbol{\varphi}$ via Γ_1 or $(\tilde{\boldsymbol{\varphi}}, \boldsymbol{N})$ via Γ_2, However in order to use sensitivity theory to relate the variation of the energy with respect to occupation numbers to the eigenvalues of the Kohn-Sham operator we focus on Γ_1.

The functional Γ_1 is not one to one and is constant on the equivalence class $\mathfrak{S}(\rho)$ of functions

$$\mathfrak{S}(\rho) = \bigcup_{1\leq\nu\leq r} \mathfrak{S}(\nu, \rho), \tag{77}$$

where the equivalence classes of constant rank, $\mathfrak{S}(\nu, \rho)$, are defined by

$$\mathfrak{S}(\nu, \rho) = \{\boldsymbol{\varphi} |\, \nu = \nu(\boldsymbol{\varphi}), \Gamma_1(\boldsymbol{\varphi}) = \rho\}. \tag{78}$$

In the following we will limit consideration to the case where $\nu(\boldsymbol{\varphi}) \geq N$ i.e. to orbitals. Each preimage set $\Gamma_1^{-1}(\rho) = \mathfrak{S}(\rho) \subset \mathfrak{S}_1$ contains, in general, both single sets of orthogonal orbitals normalized to different $\boldsymbol{N} \in \mathfrak{N}(N)$ and different sets of orthogonal functions normalized to the same $\boldsymbol{N} \in \mathfrak{N}(N)$, which corresponds to the fact that a given density can be expanded in many ways in terms of occupation numbers and orbitals.

B. Orbital functionals

One can express the total energy via a universal orbital functional, $\mathcal{F}_O$, whose value is defined by the HK energy functional, as

$$\mathcal{F}_O(\boldsymbol{\varphi}) = \mathcal{F}_{HK}(\Gamma_1(\boldsymbol{\varphi})) = \mathcal{F}_{HK}(\rho). \tag{79}$$

The orbital energy functional $\mathcal{F}_O$ can be decomposed as

$$\mathcal{F}_O(\boldsymbol{\varphi}) = \mathcal{F}_T(\boldsymbol{\varphi}) + \mathcal{F}_C(\boldsymbol{\varphi}) + \mathcal{F}_{OXC}(\boldsymbol{\varphi}), \tag{80}$$

where the orbital kinetic energy functional $\mathcal{F}_T$ is given by

$$\mathcal{F}_T(\boldsymbol{\varphi}) = \sum_{1\leq j\leq r} \langle\varphi_j | K\varphi_j\rangle, \tag{81}$$

the orbital exchange-corellation functional $\mathcal{F}_{OXC}$ (which depends on $\mathcal{F}_T$), is defined as

$$\mathcal{F}_{OXC}(\boldsymbol{\varphi}) = \mathcal{F}_{XC}(\Gamma_1(\boldsymbol{\varphi})) + \mathcal{F}_K(\Gamma_1(\boldsymbol{\varphi})) - \mathcal{F}_T(\boldsymbol{\varphi}), \tag{82}$$

and one recalls the definitions HK exchange-corellation functional

$$\mathcal{F}_{XC}(\Gamma_1(\boldsymbol{\varphi})) = \mathrm{Tr}\{V_2 D_*(\Gamma_1(\boldsymbol{\varphi}))\} - \mathcal{F}_C(\Gamma_1(\boldsymbol{\varphi})) \tag{83}$$

and the HK kinetic functional

$$\mathcal{F}_K(\Gamma_1(\boldsymbol{\varphi})) = \mathrm{Tr}\{K D_*(\Gamma_1(\boldsymbol{\varphi}))\} \tag{84}$$

that only depend on the density ρ.

One should note that:

- for fixed values of ρ the HK functionals $\mathcal{F}_{XC}(\Gamma_1(\boldsymbol{\varphi}))$ and $\mathcal{F}_K(\Gamma_1(\boldsymbol{\varphi}))$ in eqs. (83) and (84) are *constant* for all $\boldsymbol{\varphi}$ that satisfy $\rho = \Gamma_1(\boldsymbol{\varphi})$, while the orbital functionals $\mathcal{F}_T(\boldsymbol{\varphi})$ and $\mathcal{F}_{OXC}(\boldsymbol{\varphi})$ can have *different* values for values of $\boldsymbol{\varphi}$ that correspond to a fixed values of ρ

- the total energy functionals $\mathcal{F}_O(\boldsymbol{\varphi})$, and $\mathcal{F}_{HK}(\rho)$ have the same value when $\rho = \Gamma_1(\boldsymbol{\varphi}) \equiv \rho(\boldsymbol{\varphi})$

- the composition in eq. (79) defines a class of orbital functionals that have a particular form. One could define orbital functionals without reference to densities (as done in section IV B) and their form and properties would be quite different.

One can recast the minimization problem that determines the ground state energy and density of a system characterized by a local potential v in terms of orbitals as

$$E_{GS}(\mathrm{v}) = \mathcal{E}(\Gamma_1(\boldsymbol{\varphi}_{GS}), \mathrm{v}) = \min_{\boldsymbol{\varphi} \in \mathfrak{S}_1} \mathcal{E}_O(\boldsymbol{\varphi}, \mathrm{v}), \tag{85}$$

where

$$\mathcal{E}_O(\boldsymbol{\varphi}, \mathrm{v}) = \mathcal{F}_O(\boldsymbol{\varphi}) + f_{\Gamma_1(\boldsymbol{\varphi})}(\mathrm{v}) \tag{86}$$

and the feasible set $\mathfrak{S}_1$ is defined in eq. (71).

The orbital minimization problem eq. (85) can be decomposed into two sequential optimizations.

1. Varying orbitals with fixed normalization

One first varies the orbitals keeping their normalization fixed and determines the saddle point $\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\boldsymbol{\varepsilon}\left(\boldsymbol{N}\right),\mu\left(N\right),\mathrm{v}\right)$ of the Lagrangian

$$\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\boldsymbol{N},\mu,\mathrm{v}\right)=\mathcal{E}_O\left(\boldsymbol{\varphi},\mathrm{v}\right)-\sum_{1\leq j,k\leq r}\varepsilon_{jk}\left\{\left\langle\varphi_j\right|\left.\varphi_k\right\rangle-N_j\delta_{jk}\right\}+\mu\left\{\sum_{1\leq j\leq r}\left\langle\varphi_j\right|\left.\varphi_j\right\rangle-N\right\} \tag{87}$$

i.e.

$$\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\boldsymbol{\varepsilon}\left(\boldsymbol{N}\right),\mu\left(N\right),\mathrm{v}\right)=\max_{\boldsymbol{\varepsilon},\mu}\min_{\boldsymbol{\varphi}}\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\mu,\boldsymbol{N},\mathrm{v}\right), \tag{88}$$

which leads to optimal values $\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)=\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\boldsymbol{\varepsilon}\left(\boldsymbol{N}\right),\mu\left(N\right),\mathrm{v}\right)$ of the orbital energy functional for fixed values of the control parameters $\boldsymbol{N}$. The particle number constraint $\sum_{1\leq j\leq r}\left\langle\varphi_j\right|\left.\varphi_j\right\rangle=N$ is equivalent to the normalization constraint imposed on the density so that $\rho\in\mathcal{D}$.

The saddle points of $\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\boldsymbol{N},\mu,\mathrm{v}\right)$ are characterized by the necessary conditions

$$\frac{\delta\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\mu,\boldsymbol{N},\mathrm{v}\right)}{\delta\varphi_j^*}=0;\ 1\leq j\leq r, \tag{89a}$$

$$\frac{\partial\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\mu,\boldsymbol{N},\mathrm{v}\right)}{\partial\varepsilon_{jk}}=0;\ 1\leq j,k\leq r \tag{89b}$$

$$\frac{\partial\mathcal{L}_{\boldsymbol{\varphi}}\left(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\mu,\boldsymbol{N},\mathrm{v}\right)}{\partial\mu}=0 \tag{89c}$$

The condition eq. $(89a)$ implies that at an optimal point the matrix of Lagrange multipliers $\boldsymbol{\varepsilon}\left(\boldsymbol{N}\right)$ is hermitian i.e.

$$\varepsilon\left(\boldsymbol{N}\right)_{jk}=\varepsilon\left(\boldsymbol{N}\right)_{kj}^*,\quad 1\leq j,k\leq r \tag{90}$$

and

$$\frac{\delta\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)}{\delta\varphi_j^*}-\sum_{1\leq k\leq r}\left(\varepsilon_{jk}\left(\boldsymbol{N}\right)-\mu\left(N\right)\delta_{jk}\right)\left|\varphi_k\left(\boldsymbol{N}\right)\right\rangle=0;\ 1\leq j\leq r. \tag{91}$$

Noting that as $\rho=\Gamma_1\left(\boldsymbol{\varphi}\right)$ and $\mathcal{E}_O\left(\boldsymbol{\varphi},\mathrm{v}\right)\equiv\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)$ one can obtain

$$\begin{aligned}\frac{\delta\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\varphi_j^*\left(\boldsymbol{r}\right)}&=\int\frac{\delta\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho\left(\boldsymbol{r}'\right)}\frac{\delta\rho\left(\boldsymbol{r}'\right)}{\delta\varphi_j^*\left(\boldsymbol{r}\right)}d\boldsymbol{r}'\\&=\int\frac{\delta\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho\left(\boldsymbol{r}\right)}\varphi_j\left(\boldsymbol{r}'\right)\delta\left(\boldsymbol{r}-\boldsymbol{r}'\right)d\boldsymbol{r}'=\frac{\delta\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho\left(\boldsymbol{r}\right)}\varphi_j\left(\boldsymbol{r}\right)\end{aligned} \tag{92}$$

which gives

$$\frac{\delta\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho}\left|\varphi_j\right\rangle=\left|\frac{\delta\mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho}\varphi_j\right\rangle, \tag{93}$$

where the operator $\frac{\delta \mathcal{E}_O(\rho(\boldsymbol{\varphi}),\mathrm{v})}{\delta\rho}$ is a multiplication operator in the coordinate representation defined by

$$\left[\frac{\delta \mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho}\varphi_j\right](\boldsymbol{r}) = \left(\frac{\delta \mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\right),\mathrm{v}\right)}{\delta\rho\left(\boldsymbol{r}\right)}\right)\left(\varphi_j\left(\boldsymbol{r}\right)\right). \tag{94}$$

Substituting this identity back into eq. (91) one obtains for the special case of optimized orbitals, $\boldsymbol{\varphi}\left(\boldsymbol{N}\right)$, that

$$\frac{\delta \mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right)\right),\mathrm{v}\right)}{\delta\rho}\left|\varphi_j\left(\boldsymbol{N}\right)\right\rangle = \sum_{1\le k\le r}\left(\varepsilon_{jk}\left(\boldsymbol{N}\right)-\mu\left(N\right)\delta_{jk}\right)\left|\varphi_k\left(\boldsymbol{N}\right)\right\rangle, \; \varepsilon_{jk}\left(\boldsymbol{N}\right)=\varepsilon^*_{kj}\left(\boldsymbol{N}\right), \tag{95}$$

thus obtaining orbital equations corresponding to the condition eq. $(89a)$.

The eqs $(89b)$ and $(89c)$ restate the constraint conditions and as $\left\|\left|\varphi_k\left(\boldsymbol{N}\right)\right\rangle\right\| = N_k = 0, \nu+1 \le k \le r$ eqs $(87, 88, 89, 91, 95)$ *only* refer to the variables $(\varphi_1, \cdots, \varphi_\nu)$ i.e. to the occupied orbitals. One should note that eq. (95) is *not* an eigenvalue equation unless $\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)$ is invariant to unitary mixing of $\boldsymbol{\varphi}\left(\boldsymbol{N}\right)$ that produce the same density. However to this end one can invoke Harrimans theorem [10] which shows that there is always a class of IPS's, $\mathfrak{S}\left(N,\rho\right)$ (the Harriman set for densities associated to states that contain on the average N electrons) that produce a given density, so without loss of generality we can always restrict the domain of Γ_1 to $\mathfrak{S}\left(N,\rho\right)$ and replace eq. (95) by

$$\frac{\delta \mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right)\right),\mathrm{v}\right)}{\delta\rho}\left|\varphi_j\left(\boldsymbol{N}\right)\right\rangle = \left(\varepsilon_{jj}\left(\boldsymbol{N}\right)-\mu\left(N\right)\right)\left|\varphi_j\left(\boldsymbol{N}\right)\right\rangle; \quad 1\le j\le N \tag{96}$$

as $\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\iota}_N\right),\mathrm{v}\right)$, where

$$\boldsymbol{\iota}_N = \left(\overbrace{1,\cdots,1}^{N},\overbrace{0,\cdots,0}^{r-N}\right), \tag{97}$$

is inavariant to unitary mixing of $\boldsymbol{\varphi}\left(\boldsymbol{\iota}_N\right) \in \mathfrak{S}\left(N,\rho\right)$.

From sensitivity theory the rate of change of the orbital energy functional with respect to changes in occupation numbers, which in this case are the control parameters $\boldsymbol{N}$, are

$$\frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)}{\partial N_j} = \varepsilon_{jj}\left(\boldsymbol{N}\right)-\mu\left(N\right); 1\le j\le\nu, \tag{98}$$

which is Janaks Theorem for *any* fixed set $\boldsymbol{N}$ of occupation numbers and optimized orbitals $\boldsymbol{\varphi}\left(\boldsymbol{N}\right)$ normalized to these values. If we take expansion eq. (95) into account we can obtain these relationships as expectation values of the operator $\frac{\delta \mathcal{E}_O(\rho(\boldsymbol{\varphi}(\boldsymbol{N})),\mathrm{v})}{\delta\rho} - \mu I$, i.e.

$$\frac{\partial \mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right)\right),\mathrm{v}\right)}{\partial N_j} = \left\langle \tilde{\varphi}_j\left(\boldsymbol{N}\right)\middle|\left\{\frac{\delta \mathcal{E}_O\left(\rho\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right)\right),\mathrm{v}\right)}{\delta\rho}-\mu\left(N\right)I\right\}\tilde{\varphi}_j\left(\boldsymbol{N}\right)\right\rangle = \varepsilon_{jj}\left(\boldsymbol{N}\right)-\mu\left(N\right), \tag{99}$$

where

$$\left|\tilde{\varphi}_j\left(\boldsymbol{N}\right)\right\rangle = \frac{\left|\varphi_j\left(\boldsymbol{N}\right)\right\rangle}{\left(\left\langle\varphi_j\left(\boldsymbol{N}\right)\right|\left.\varphi_j\left(\boldsymbol{N}\right)\right\rangle\right)^{\frac{1}{2}}}. \tag{100}$$

By considering the sensitivity of the energy to changes of the average number of electrons in the system one obtains

$$\frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)}{\partial N} = \mu\left(N\right) \tag{101}$$

and $\mu\left(N\right)$ can be interpreted as the chemical potential of the interacting molecular system.

2. Varying the normalization

In the second optimization we can restrict attention to control parameters, $\boldsymbol{N}$, such that $\nu\left(\boldsymbol{N}\right) \geq N$ and $0 \leq N_j \leq 1, 1 \leq j \leq r$, i.e. that can be interpretted as occupation numbers. There one finds minima of $\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}\right),\mathrm{v}\right) \equiv \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\left(\boldsymbol{\theta}\right)\right),\mathrm{v}\right)$ with respect to a constrained variation of the normalization control parameters $\boldsymbol{N} = N\left(\boldsymbol{\theta}\right)$ parameterized by $\boldsymbol{\theta}$, where

$$0 \leq \theta_j \leq \frac{\pi}{2}; \quad 1 \leq j \leq r$$

for a fixed N that are constrained by $\sum_{1\leq j\leq r} N_j = \sum_{1\leq j\leq r} \sin\theta_j = N$. Stationary points $\boldsymbol{\theta}_{\#}\left(N\right)$ are determined by

$$\mathcal{L}_{\boldsymbol{\theta}}\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),N,\eta\left(N\right),\mathrm{v}\right) = \max_{\boldsymbol{\eta}}\min_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\theta}}\left(\boldsymbol{\varphi},N,\eta,\mathrm{v}\right) \tag{102}$$

the Lagrangian

$$\mathcal{L}_{\boldsymbol{\theta}}\left(\boldsymbol{\varphi},N,\eta,\mathrm{v}\right) = \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}\right),\mathrm{v}\right) + \eta\left\{\sum_{1\leq j\leq r} \sin\theta_j - N\right\} \tag{103}$$

where

$$\begin{aligned} \frac{\partial \mathcal{L}_{\boldsymbol{\theta}}\left(\boldsymbol{\varphi},N,\eta,\mathrm{v}\right)}{\partial\theta_j} &= 0; 1 \leq j \leq r \\ \frac{\partial \mathcal{L}_{\boldsymbol{\theta}}\left(\boldsymbol{\varphi},N,\eta,\mathrm{v}\right)}{\partial\eta} &= 0. \end{aligned} \tag{104}$$

The derivatives with respect to $\boldsymbol{\theta}$ are given by the chain rule

$$\frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}\left(N\right)\right),\mathrm{v}\right)}{\partial\theta_j} = \sum_{1\leq k\leq r} \frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)}{\partial N_k}\frac{\partial N_k}{\partial\theta_j} = \frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{N}\right),\mathrm{v}\right)}{\partial N_j}\cos\theta_j; \quad 1 \leq j \leq r \tag{105}$$

and

$$\frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\mathbf{N}\right),\mathrm{v}\right)}{\partial N_j}=\varepsilon_{jj}\left(\mathbf{N}\left(\boldsymbol{\theta}\right)\right)-\mu\left(N\right). \tag{106}$$

Giving

$$\frac{\partial \mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),\mathrm{v}\right)}{\partial \theta_j}=\left\{\varepsilon_{jj}\left(\mathbf{N}\left(\boldsymbol{\theta}_{\#}\right)\right)-\mu\left(N\right)\right\}\cos\theta_{\# j}=0. \tag{107}$$

Thus

$$\varepsilon_{jj}\left(\mathbf{N}\left(\boldsymbol{\theta}_{\#}\right)\right)-\mu\left(N\right)=0 \text{ and/or } \left\{\cos\theta_{\# j}=0\Leftrightarrow\theta_{\# j}=\frac{\pi}{2}\right\};\quad 1\leq j\leq r. \tag{108}$$

If $\theta_j=\frac{\pi}{2}$ $\left(N_j\left(\boldsymbol{\theta}_{\#}\right)=1\right)$ then ε_{jj} can have any value (even $\mu\left(N\right)$), while if $\theta_{\# j}\neq\frac{\pi}{2}$ (which corresponds to non integer occupation) then $\varepsilon_{jj}=\mu\left(N\right)$ i.e. the Lagrange parameters $\left\{\varepsilon_{jj}\right\}$ are degenerate. If $\varepsilon_{jj}=\mu\left(N\right)$ then $\theta_{\# j}$ does not have to have the value $\frac{\pi}{2}$ though it can. Since the inception of the KS approach to DFT there has been an extensive discussion of the physical meaning of the eigenvalues of the KS operator and their relationship to the Fermi energy and chemical potential.

The ground state energy $E\left(\rho_{GS},\mathrm{v}\right)=\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),\mathrm{v}\right)$ and the ground state density can then be expanded as

$$\rho_{GS}\left(\boldsymbol{r}\right)=\sum_{1\leq j\leq r}\left|\left[\varphi_j\left(\boldsymbol{\theta}_{\#}\right)\right]\left(\boldsymbol{r}\right)\right|^2. \tag{109}$$

The expansion eq. (109) is very *non* unique in that any set $\boldsymbol{\varphi}\left(\boldsymbol{\theta}'_{\#}\right)\in\mathfrak{S}\left(\rho_{GS}\right)$, where $\boldsymbol{\theta}'_{\#}$ satisfies eq. (107) produces ρ_{GS} and the same value of the energy functional i.e. $\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}'_{\#}\right),\mathrm{v}\right)=\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),\mathrm{v}\right)$.

C. The Kohn-Sham Density Functional

As pointed out in the last subsection the minima of $\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),\mathrm{v}\right)$,

$$\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),\mathrm{v}\right)=\min_{\boldsymbol{\varphi}\in\mathfrak{S}_1}\mathcal{E}_O\left(\boldsymbol{\varphi},\mathrm{v}\right) \tag{110}$$

are highly degenerate e.g.

$$\mathcal{E}_O\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}_{\#}\right),\mathrm{v}\right)=const.\,\forall\,\boldsymbol{\varphi}\in\mathfrak{S}\left(\rho_{GS}\right). \tag{111}$$

This degeneracy is due to the singularity of the map Γ_1 which can be removed by constraining $\boldsymbol{\varphi}$ to be a specific representative of the equivalence classes $\mathfrak{S}\left(\rho\right)$. One way to obtain such a representative is to consider the orbitals formed by the NOsand occupation numbers of the

FORDO of the N-electron ground state $D_0(\rho) = A_0(\rho)^2$ of a model independent particle N-electron system, that has a fixed density ρ. This corresponds to finding the minimizers of the universal kinetic energy functional K_U

$$K_U(A) = \mathrm{Tr}\left\{KA^2\right\}, \tag{112}$$

subject to the constraint

$$\Xi_2(A) = \left(A\left|\widehat{\Phi^\dagger\Phi}A\right.\right) = \rho,\ A \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right). \tag{113}$$

The constrained minima $K_U(D_0(\rho)) = \mathrm{Tr}\{KD_0(\rho)\} = \mathrm{Tr}\left\{KA_0(\rho)^2\right\}$ is given as an unconstrained saddle point $(A_0(\rho), \lambda_0(\rho))$,

$$\mathcal{L}_K(A_0(\rho), \lambda_0(\rho), \rho) = \max_{\lambda}\min_{A}\mathcal{L}_K(A, \lambda, \rho) \tag{114}$$

of the Lagrangian

$$\mathcal{L}_K(A, \lambda, \rho) = K_U(A) - \Omega_A(\lambda) + \langle\lambda|\,\rho\rangle = \left(A\left|\left\{\left\{\widehat{K} - \widehat{\Omega(\lambda)}\right\} + \langle\lambda|\,\rho\rangle\,\hat{I}\right\}A\right.\right). \tag{115}$$

If a bound state minimum exists for some local one body potential $\lambda_0(\rho)$ (which is also the Lagrange parameter function) one has

$$\left\{\widehat{K} - \Omega\widehat{(\lambda_0(\rho))}\right\}|A_0(\rho)) = 0, \tag{116}$$

where $A_0(\rho) \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$. This gives a stationary state $D_0(\rho) = A_0(\rho)^2$ of the independent particle system with Hamiltonian $\{K - \Omega(\lambda_0(\rho))\}$ i.e.

$$[\{K - \Omega(\lambda_0(\rho))\}, D_0(\rho)] = 0, \tag{117}$$

which in turn defines a FORDO $D_0^1(\rho)$ given by (see Appendix A)

$$D_0^1(\rho) = NL_N^1(D_0(\rho)). \tag{118}$$

As $K - \Omega(\lambda_0(\rho))$ is an one-electron operator the state $D_0(\rho)$ must be a mixture of IPSsor a pure IPS and thus determined by the eigenvalues $\{N_{0j}(\rho)\}$ and eigenvectors $\left\{\left|\widetilde{\varphi_{0j}(\rho)}\right\rangle\right\}$ of $D_0^1(\rho)$. If $D_0(\rho)$ is a pure IPS then the occupation numbers $\{N_{0j}(\rho)\}$ are equal to either 1 or 0 (i.e. N ones and $r - N$ zeroes), while if it is a ensemble state than there will be some fractional values and the number of zero values will be less than $r-N$. Expanding $D_0^1(\rho)$ in terms of normalized NOs$\{|\widetilde{\varphi_{0j}}\rangle|\, 1\leq j\leq r\}$ gives

$$D_0^1(\rho)=\sum_{1\leq j\leq r} N_{0j}(\rho)\left|\widetilde{\varphi_{0j}(\rho)}\right\rangle\left\langle\widetilde{\varphi_{0j}(\rho)}\right|, \tag{119}$$

(where in contrast to eq. $(70b)$ $r\geq N$), which after renormalizing the NOsto the occupation numbers i.e.

$$|\varphi_{0j}(\rho)\rangle=N_{0j}(\rho)^{\frac{1}{2}}\left|\widetilde{\varphi_{0j}(\rho)}\right\rangle, 1\leq j\leq r \tag{120}$$

results in

$$\rho(\boldsymbol{r})=\sum_{1\leq j\leq r}\left|[\varphi_{0j}(\rho)](\boldsymbol{r})\right|^2. \tag{121}$$

If the constrained minimum of K_U is non degenerate then there is a unique correspondence, up to unitary mixing of degenerate eigenvectors of $D_0^1(\rho)$, between ρ and $\{\boldsymbol{\varphi}_0\}$, thus modulo this unitary mixing one can define an inverse functional Γ_0^{-1} as

$$\boldsymbol{\varphi}_0=\Gamma_0^{-1}(\rho). \tag{122}$$

If the constrained minimum of K_U is degenerate, e.g. corresponding to FORDOs$\{D_{0\kappa}^1(\rho)|\, 1\leq\kappa\leq m\}$ of the stationary states $\{D_{0\kappa}(\rho)|\, 1\leq\kappa\leq m\}$ of $K-\Omega(\lambda_0(\rho))$ then as

$$\left[\{K-\Omega(\lambda_0(\rho))\}, D_{0\kappa}(\rho)\right]=\left[\{K-\Omega(\lambda_0(\rho))\}, D_{0\kappa}^1(\rho)\right]=0, \tag{123}$$

the space of degenerate states corresponds to the same set of orbitals $\{|\boldsymbol{\varphi}_0(\rho)\rangle\}$ but normalized to different occupation numbers $\{\boldsymbol{N}_0(\rho)\}$, where $\boldsymbol{N}_0(\rho)$ are convex combinations of $\{\boldsymbol{N}_{0\kappa}(\rho)|\, 1\leq\kappa\leq m\}$. Hence if degeneracies occur these are reflected in multiple optimal values for $\boldsymbol{N}_0$ for fixed $\{\boldsymbol{\varphi}_0(\rho)\}$, some of the $\{N_{0j}|\, 1\leq j\leq r\}$ can have non integer values even for degenerate pure states.

The minimal values of the universal kinetic functional $K_U(D_0(\rho))$ produce a kinetic energy functional $\mathcal{F}_T$ defined on the representatives $\{\boldsymbol{\varphi}_0\}$ of the equivalence classes $\mathfrak{S}(\rho)$ whose values are given by

$$\mathcal{F}_T(\boldsymbol{\varphi}_0)=K_U(D_0(\Gamma_1(\boldsymbol{\varphi}_0))). \tag{124}$$

The value of $\mathcal{F}_T$ is constant on the set of orbitals formed from the unitary mixing and the convex combinations of degenerate states just described. This allows one to define the

Kohn-Sham functional $\mathcal{F}_{KS}(\boldsymbol{\varphi}_0)$ as a specific decomposition of the HK energy functional $\mathcal{F}_{HK}(\rho)$ in terms of orbitals as

$$\mathcal{F}_{KS}(\boldsymbol{\varphi}_0) = \mathcal{F}_T(\boldsymbol{\varphi}_0) + \mathcal{F}_C(\boldsymbol{\varphi}_0) + \mathcal{F}_{KSXC}(\boldsymbol{\varphi}_0), \tag{125}$$

where

$$\mathcal{F}_{KS}(\boldsymbol{\varphi}_0) = \mathcal{F}_{HK}(\Gamma_1(\boldsymbol{\varphi}_0)). \tag{126}$$

The relationships described in subsection III B derived for stationary points $\boldsymbol{\varphi}(\boldsymbol{N})$ are valid for any points contained in the equivalence classes that these points belong to, in particular $\boldsymbol{\varphi}_0(\boldsymbol{N}_0)$. Eq. (95), is now an equation for determining the KS orbitals $\boldsymbol{\varphi}_0(\boldsymbol{N})$ and the operator introduced there can be identified as the KS operator. If the state $D_0(\rho)$ is non degenerate and pure (therefore has only occupation numbers equal to one or zero) then one can choose a set of orbitals that diagonalize this operator as the energy $\mathcal{E}_O(\boldsymbol{\varphi}_0(\boldsymbol{N}_0), \mathrm{v})$ is invariant to unitary mixing of occupied orbitals, which is the standard way of producing a canonical set of occupied orbitals.

The reference N-electron IPS, $D_0(\boldsymbol{N}_0)$ formed from the NOs$\boldsymbol{\varphi}_0$ has two energies associated with it. One, $E_0(\boldsymbol{\varphi}_0(\boldsymbol{N}_0))$, is the ground state energy of the Hamiltonian

$$H_0 = K + \mathrm{v}, \tag{127}$$

which can be expressed as

$$E_0(\boldsymbol{\varphi}_0(\boldsymbol{N}_0)) = \sum_{1 \le j \le r} \langle \tilde{\varphi}_{0j} | (K + \mathrm{v}) \, \tilde{\varphi}_{0j} \rangle \, N_{0j}. \tag{128}$$

The other is the energy of the Hamiltonian

$$H = H_0 + V_2 \tag{129}$$

wrt the IPS $D_0(\boldsymbol{N}_0)$ given by

$$E_{IP}(D_0(\boldsymbol{N}_0)) = \mathrm{Tr}\{H D_0(\boldsymbol{N}_0)\}. \tag{130}$$

It should be noted that this value of the energy functional E_{IP} is *not* optimized in any way and is *not* related in any simple fashion to the energies $\mathcal{E}_O(\varphi(\mathbf{N_0}), \mathrm{v})$.

The preceding relationships are equally for orbitals $\boldsymbol{\varphi}_0(\boldsymbol{N}_0(\boldsymbol{\theta}_\#(N)))$ that produce the ground state density ρ_{GS} and energy $\mathcal{E}_O(\boldsymbol{\varphi}_0(\boldsymbol{N}(\boldsymbol{\theta}_\#(N))), \mathrm{v})$ and the classical Janaks Theorem is a special case of eq. (98)

$$\frac{\partial \mathcal{E}_O(\boldsymbol{\varphi}_0(\boldsymbol{N}_0(\boldsymbol{\theta}_\#(N))), \mathrm{v})}{\partial N_j} \equiv \frac{\partial \mathcal{E}_{KS}(\boldsymbol{\varphi}_0(\boldsymbol{N}_0(\boldsymbol{\theta}_\#(N))), \mathrm{v})}{\partial N_j} = \varepsilon_{jj}(\boldsymbol{\varphi}_0(\boldsymbol{N}_0(\boldsymbol{\theta}_\#(N)))), \tag{131}$$

where we have constrained the orbitals and occupation numbers to be the ones that minimize the universal kinetic energy, K_U, thus producing a model IPS. One must, however, note that the Lagrange parameters $\{\varepsilon_{jj}(\boldsymbol{\varphi}_0(\boldsymbol{N}_0(\boldsymbol{\theta}_\#(N))))\}$ in this equation *are not* associated with the minimization of the universal kinetic energy.

The KS energy functional is defined in terms of orbitals and occupation numbers and is implicitly defined by the preceding constrained optimization procedures, that factor the functional through the density. If one did not consider this factorization and directly optimized the orbitals and occupation numbers one would determine the same ground state, but not need to introduce a model IPS and orbitals $\boldsymbol{\varphi}_0(\boldsymbol{N}_0)$. In the next section we compare and contrast the relationships obtained for these functionals with ones obtained for the explicitly defined AGP functional. These functionals are defined on the space of correlated states formed by the non linear manifold of AGP states, where the energy has been expressed as an explicit functional of GSOsand parameters that determine occupation numbers [3] The minimization of this explicit functional can then be compared to the minimization of the implicit KS functional restricted to the AGP manifold.

IV. AGP ENERGY FUNCTIONALS

We briefly review some of the results found in [3] describing AGP states and their associated energy functionals. In [3] orbital and occupation number optimization equations were derived and it was displayed how the N-electron total energy could be expressed in terms of ionization potentials, kinetic energies and eigenvalues of a generalized Fock operator associated with GSO's. In contrast to the KS approach, the occupation numbers were not constrained to refer to IPS's, but allowed to vary over the much larger domain of states with doubly degenerate state occupation numbers. It has been argued while this set does not contain all possible states it contains a large percentage of states important for molecular systems [**?**]. The density matrix energy functional in DMFT allows unrestricted variation of occupation numbers (and thus all states), but unlike the AGP functional (but in common with the HK and KS density functionals) it is invariant to the phases of the GSOsi.e. the group $U(1)$.

A. Canonical expansion of an AGP state

The $2M$-electron AGP state, $\left|g^M\right\rangle$, is the M^{th} antisymmetric power of the 2-electron state described by a normalized geminal

$$|g\rangle = \sum_{1\le j\le s} |\varphi_j\varphi_{j+s}\rangle = \sum_{1\le j\le s} n_j^{\frac{1}{2}} |\tilde{\varphi}_j\tilde{\varphi}_{j+s}\rangle\,, \tag{132}$$

given by

$$\left|g^M\right\rangle = (S_M)^{-\frac{1}{2}} \sum_{1\le j_1<\cdots<j_M\le s} n_{j_1}^{\frac{1}{2}}\cdots n_{j_s}^{\frac{1}{2}} |\tilde{\varphi}_{j_1}\tilde{\varphi}_{j_1+s}\cdots\tilde{\varphi}_{j_M}\tilde{\varphi}_{j_M+s}\rangle = (S_M)^{-\frac{1}{2}} \sum_{1\le j_1<\cdots<j_M\le s} |\varphi_{j_1}\varphi_{j_1+s}\cdots\varphi_{j_M}\varphi_{j_M+} \tag{133}$$

wherer $\left\{ n_j^{\frac{1}{2}} \middle| 1\le j\le s\right\}$ are the positive square roots of the occupation numbers

$$\Omega = \left\{ n_j \middle| 0\le n_j\le 1; \sum_{1\le j\le s} n_j = 1; 1\le j\le s\right\} \tag{134}$$

of the FORDO of the two electron AGP state density operator, $\{|\tilde{\varphi}_j\rangle|\, 1\le j\le 2s\}$ are the *orthonormal* Canonical General Spin Orbitals (CGSO's) of $|g\rangle$ and $\{|\varphi_j\rangle|\, 1\le j\le 2s\}$ are the *orthogonal* CGSOsnormalized to $\boldsymbol{n}$ i.e.

$$|\varphi_j\rangle = n_j^{\frac{1}{2}} |\tilde{\varphi}_j\rangle\,, 1\le j\le 2s. \tag{135}$$

The normalization factor, $(S_M)^{-\frac{1}{2}}$, is given in terms of the elementary symmetric function

$$S_M = \sum_{1\le j_1<\cdots<j_M\le s} n_{j_1}\cdots n_{j_M}. \tag{136}$$

It is evident from eq. (132) that the geminal and geminal power state are at least invariant to the group $SU(2)$, that is formed by 2×2 *special* unitary transformations of the paired CGSOs$\{|\tilde{\varphi}_j\rangle\,, |\tilde{\varphi}_{j+s}\rangle|\, 1\le j\le s\}$, while if degeneracies exist amongst the $\{n_j|\, 1\le j\le s\}$ this invariance group is larger [11]. The canonical CGSOsare unique up to transformations belonging this invariance group.

B. AGP Energy Functionals

The energy of the AGP state can be expressed as a functional $\mathcal{E}_{AGP}$ of $\boldsymbol{\varphi}(\boldsymbol{n})$ by

$$\begin{aligned}\mathcal{E}_{AGP}(\boldsymbol{\varphi}(\boldsymbol{n})) &= \frac{\mathcal{H}(\boldsymbol{\varphi}(\boldsymbol{n}))}{S_M(\boldsymbol{n})} \\ &= \frac{1}{S_M(\boldsymbol{n})}\left\{\sum_{1\le j\le s} n_j S_{M-1}(\jmath)\left\{\langle\tilde{\varphi}_j|h_1\,\tilde{\varphi}_j\rangle + \langle\tilde{\varphi}_{j+s}|h\,\tilde{\varphi}_{j+s}\rangle + \langle\tilde{\varphi}_j\tilde{\varphi}_{j+s}||\tilde{\varphi}_j\tilde{\varphi}_{j+s}\rangle\right\}\right. \\ &\left. + \sum_{1\le j,k\le s} n_j^{\frac{1}{2}} n_k^{\frac{1}{2}} S_{M-1}(\jmath k)\langle\tilde{\varphi}_j\tilde{\varphi}_{j+s}||\tilde{\varphi}_k\tilde{\varphi}_{k+s}\rangle + \sum_{1\le j<k\le s} n_i n_j S_{M-2}(\jmath k)\langle\tilde{\varphi}_j\tilde{\varphi}_k|||\tilde{\varphi}_j\tilde{\varphi}_k\rangle\right\}.\end{aligned} \tag{137}$$

C. AGP Lagrangians

Using the same philosophy for optimization as discussed for orbital functionals in the preceding sections one can minimize the energy of the $2M$-electron state over the manifold of AGP states using a nested two step process:

1. Keeping the normalization $\boldsymbol{n}$ of the GSOs$\boldsymbol{\varphi}$ fixed one finds the constrained minimum $\mathcal{E}_{AGP}(\boldsymbol{\varphi}(\boldsymbol{n}))$ of the functional $\mathcal{E}_{AGP}(\boldsymbol{\varphi})$ for the specified values of the control parameters $\boldsymbol{n}$ i.e.

$$\mathcal{E}_{AGP}(\boldsymbol{\varphi}(\boldsymbol{n})) = \min_{\boldsymbol{\varphi}\in\Lambda(\boldsymbol{n})} \mathcal{E}_{AGP}(\boldsymbol{\varphi}), \tag{138}$$

where

$$\Lambda(\boldsymbol{n}) = \left\{\boldsymbol{\varphi}\,\middle|\,\langle\varphi_j|\,\varphi_k\rangle = n_j\delta_{jk};\ \sum_{1\le j\le s} n_j = 1\right\}.$$

This is accomplished by finding the unconstrained saddle points $\mathcal{L}(\boldsymbol{\varphi}(\boldsymbol{n}), \boldsymbol{\varepsilon}(\boldsymbol{n}), \eta)$ of the Lagrangian

$$\mathcal{L}(\boldsymbol{\varphi},\boldsymbol{\varepsilon}) = \mathcal{E}_{AGP}(\boldsymbol{\varphi}) - \sum_{1\le j,k\le 2s}\varepsilon_{kj}\left(\langle\varphi_j|\,\varphi_k\rangle - n_j\delta_{jk}\right) + \eta\left\{\sum_{1\le j\le 2s}\langle\varphi_j|\,\varphi_j\rangle - 2\right\}, \tag{139}$$

i.e.

$$\mathcal{L}(\boldsymbol{\varphi}(\boldsymbol{n}), \boldsymbol{\varepsilon}(\boldsymbol{n}), \eta) = \max_{\boldsymbol{\varepsilon},\eta.}\min_{\boldsymbol{\varphi}} \mathcal{L}(\boldsymbol{\varphi},\boldsymbol{\varepsilon},\eta) \tag{140}$$

The corresponding stationarity conditions are

$$\frac{\delta \mathcal{L}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right), \boldsymbol{\varepsilon}\left(\boldsymbol{n}\right), \eta\right)}{\delta \boldsymbol{\varphi}} = \mathbf{0} \tag{141a}$$

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right), \boldsymbol{\varepsilon}\left(\boldsymbol{n}\right), \eta\right)}{\partial \boldsymbol{\varepsilon}} = \mathbf{0} \tag{141b}$$

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right), \boldsymbol{\varepsilon}\left(\boldsymbol{n}\right), \eta\right)}{\partial \eta} = \mathbf{0} \tag{141c}$$

and eq. (141a) gives

$$\frac{\delta \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\delta \varphi_j} = \sum_{1 \leq k \leq 2s} \varepsilon(\boldsymbol{n})_{jk} \left|\varphi_k(\boldsymbol{n})\right\rangle, \tag{142}$$

which can be expressed [3] in terms of a generalized Fock operator as

$$F\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)\left|\varphi_j\left(\boldsymbol{n}\right)\right\rangle = \sum_{1 \leq k \leq 2s} \varepsilon(\boldsymbol{n})_{jk} \left|\varphi_k(\boldsymbol{n})\right\rangle. \tag{143}$$

Sensitivity theory gives

$$\frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial n_j} = \varepsilon_{jj}\left(\boldsymbol{n}\right).$$

2. Varying $\mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)$ over the normalization control parameters $\boldsymbol{n} \in \Omega$ to obtain the minimum $\mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}_{\#}\right)\right)$ i.e.

$$\mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}_{\#}\right)\right) = \min_{\boldsymbol{n} \in \Omega} \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right). \tag{144}$$

The optimization eq. (144) can be made unconstrained by expressing $\boldsymbol{n}$ in terms of the variable $\boldsymbol{\theta}$ as

$$n_j\left(\boldsymbol{\theta}\right) = \frac{\cos^2\theta_j}{\sum_{1 \leq j \leq s} \cos^2\theta_j} \equiv \frac{\cos^2\theta_j}{\mathcal{N}_n\left(\boldsymbol{\theta}\right)}; 0 \leq \theta_j \leq \frac{\pi}{2} \tag{145}$$

⌈

$$\sum_{1 \leq j \leq s} \frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial n_j} \frac{\partial n_j}{\partial \theta_k} = 0; 1 \leq k \leq s. \tag{146}$$

Using eq.(145) and observing that

$$\frac{\partial n_j}{\partial \theta_k} = \frac{\cos^2\theta_j \sin 2\theta_k}{\mathcal{N}\left(\boldsymbol{\theta}\right)^2} - \frac{\sin 2\theta_k \delta_{jk}}{\mathcal{N}\left(\boldsymbol{\theta}\right)} = \frac{\sin 2\theta_k}{\mathcal{N}\left(\boldsymbol{\theta}\right)} \left\{\frac{\cos^2\theta_j}{\mathcal{N}\left(\boldsymbol{\theta}\right)} - \delta_{jk}\right\} \tag{147}$$

and

$$\frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\left(\boldsymbol{\theta}\right)\right)\right)}{\partial n_j} = \varepsilon_{jj}\left(\boldsymbol{n}\left(\boldsymbol{\theta}\right)\right) \tag{148}$$

this can be developed to give

$$\frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{\theta}\right)\right)}{\partial \theta_k} = \sum_{1\leq j\leq s} \varepsilon_{jj}\left(\boldsymbol{\theta}\right) \frac{\sin 2\theta_k}{\mathcal{N}\left(\boldsymbol{\theta}\right)} \left\{ \frac{\cos^2\theta_j}{\mathcal{N}\left(\boldsymbol{\theta}\right)} - \delta_{jk} \right\} = 0$$

$$\Rightarrow \sum_{1\leq j\leq s} \varepsilon_{jj}\left(\boldsymbol{\theta}\right) \frac{\cos^2\theta_j \sin 2\theta_k}{\mathcal{N}\left(\boldsymbol{\theta}\right)^2} = \varepsilon_{kk}\left(\boldsymbol{\theta}\right)\sin 2\theta_k \tag{149}$$

⌋

We are interested in the dependence on the occupation numbers of the $2M$-electron state which we can obtain by using the chain rule

$$\frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial N_j} = \sum_{1\leq k\leq 2s} \frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial n_k}\frac{\partial n_k}{\partial N_j}$$

$$\Rightarrow \frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial N_j} = \sum_{1\leq k\leq 2s} \frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial n_k}\frac{\partial n_k}{\partial N_j} = \sum_{1\leq k\leq 2s} \frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial n_k}\left(\frac{\partial N_k}{\partial n_j}\right)^{-1} \tag{150}$$

i.e.

$$\frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial \boldsymbol{N}} = \frac{\partial \mathcal{E}_{AGP}\left(\boldsymbol{\varphi}\left(\boldsymbol{n}\right)\right)}{\partial \boldsymbol{n}}\left(\frac{\partial \boldsymbol{N}}{\partial \boldsymbol{n}}\right)^{-1}. \tag{151}$$

The Jacobian $\frac{\partial \boldsymbol{N}}{\partial \boldsymbol{n}}$ is nonsingular as the occupation numbers of the AGP state are explicitly given in a $1-1$ fashion by []

$$\boldsymbol{N} = \mathcal{N}\left(\boldsymbol{n}\right), \tag{152}$$

where

$$N_j = \mathcal{N}_j\left(\boldsymbol{n}\right) = n_j\frac{S_{M-1}\left(\jmath\right)}{S_M} = n_j\frac{\partial \ln S_M}{\partial n_j}; \quad 1\leq j\leq s. \tag{153}$$

The matrix elements of the Jacobian $\left(\frac{\partial \boldsymbol{N}}{\partial \boldsymbol{n}}\right)$ are

$$\frac{\partial N_k}{\partial n_j} = \frac{\partial}{\partial n_j}\left\{n_k\frac{\partial \ln S_M}{\partial n_k}\right\} = \delta_{jk}\frac{\partial \ln S_M}{\partial n_k} + n_k\frac{\partial^2 \ln S_M}{\partial n_j \partial n_k}$$

$$= \delta_{jk}\frac{S_{M-1}\left(\jmath\right)}{S_M} - n_k\frac{S_{M-1}\left(\jmath\right)S_{M-1}\left(k\right)}{S_M^2} + n_k\frac{S_{M-2}\left(\jmath k\right)}{S_M}. \tag{154}$$

V. DISCUSSION

APPENDIX A: STATE DENSITY, REDUCED DENSITY OPERATORS AND OPERATOR SPACES

1. State Density Operators and Operator Spaces

The convex set S_N of normalized N-electron density operators D is defined as

$$S_N = \left\{ D \,\middle|\, D \in \mathcal{B}_{1+}\left(\mathcal{H}^N\right); \mathrm{Tr}\, D = 1 \right\}, \tag{A1}$$

where $\mathcal{B}_{1+}\left(\mathcal{H}^N\right)$ is the cone (see below) of positive trace class operators, defined as

$$\mathcal{B}_{1+}\left(\mathcal{H}^N\right) = \left\{ D \in \mathcal{B}_1\left(\mathcal{H}^N\right) \middle| \langle \Psi \,| D\Psi \rangle \geq 0 \quad \forall \Psi \in \mathcal{H}^N \right\}, \tag{A2}$$

where $\mathcal{B}_1\left(\mathcal{H}^N\right)$ is the space of trace class operators defined as

$$\mathcal{B}_1\left(\mathcal{H}^N\right) = \left\{ X \middle| \, \mathrm{Tr} \left\{ \left(X^\dagger X\right)^{\frac{1}{2}} \right\} < \infty \right\}, \tag{A3}$$

which is contained in the space of bounded N-electron operators $\mathcal{B}\left(\mathcal{H}^N\right)$ defined as

$$\mathcal{B}\left(\mathcal{H}^N\right) = \left\{ X \middle| \sup_{\substack{\Psi \in \mathcal{H}^N \\ \|\Psi\|=1}} \left\{ \left\langle \Psi \,\middle|\, X^\dagger X \Psi \right\rangle \right\} < \infty = \sup_{\substack{\Psi \in \mathcal{H}^N \\ \|\Psi\|=1}} \left\{ \|X\Psi\| \right\} < \infty \right. \tag{A4}$$

$$\left. = \sup_{\substack{\Psi, \Phi \in \mathcal{H}^N \\ \|\Psi\|=1, \|\Phi\|=1}} \left\{ \left\langle \Psi \,\middle|\, X^\dagger X \Phi \right\rangle \right\} < \infty \right\}. \tag{A5}$$

It can be shown that if one considers bounded self-adjoint operators then

$$\|X\| = \sup_{\substack{\Psi \in \mathcal{H}^N \\ \|\Psi\|=1}} \left\{ \left| \langle \Psi \,| X\Psi \rangle \right| \right\} = r\left(X\right) < \infty, \tag{A6}$$

where $r\left(X\right)$ is the spectral radius of X.

The convex set property of S_N is explicitly displayed by

$$S_N = \left\{ \sum_{1 \leq j \leq m} \alpha_j D_j \middle| \sum_{1 \leq j \leq m} \alpha_j = 1; 0 \leq \alpha_j \leq 1, D_j \in S_N, 1 \leq j \leq m \right\}. \tag{A7}$$

and a set C is a cone if $X \in C \Leftrightarrow \alpha X \in C \; \forall \alpha > 0$, which is satisfied by $\mathcal{B}_{1+}\left(\mathcal{H}^N\right)$.

The spaces $\mathcal{B}\left(\mathcal{H}^N\right)$ and $\mathcal{B}_1\left(\mathcal{H}^N\right)$ are Banach spaces with norms defined by

$$\|X\| = \left[\sup_{\substack{\Psi\in\mathcal{H}^N \\ \Psi\neq 0}} \left\{ \frac{\left\langle \Psi \left| X^\dagger X \Psi \right. \right\rangle}{\left\langle \Psi \left| \Psi \right. \right\rangle} \right\}\right]^{\frac{1}{2}} = \sup_{\substack{\Phi,\Psi\in\mathcal{H}^N \\ \Phi,\Psi\neq 0}} \left\{ \frac{\left|\left\langle \Phi \left| X\Psi \right. \right\rangle\right|}{\|\Phi\| \, \|\Psi\|} \right\} \tag{A8}$$

and

$$\|X\|_1 = \mathrm{Tr}\left\{ \left(X^\dagger X\right)^{\frac{1}{2}} \right\} \tag{A9}$$

respectively. A closely related space that we have utilized is the space of Hilbert Schimdt operators $\mathcal{B}_2\left(\mathcal{H}^N\right)$ defined as

$$\mathcal{B}_2\left(\mathcal{H}^N\right) = \left\{ X \left| \mathrm{Tr}\left\{X^\dagger X\right\} < \infty \right. \right\}, \tag{A10}$$

which is a Hilbert space with inner product given by

$$\left(X \left| Y \right. \right) = \mathrm{Tr}\left\{X^\dagger Y\right\} \tag{A11}$$

and norm by

$$\|X\|_2 = \left(\mathrm{Tr}\left\{X^\dagger X\right\}\right)^{\frac{1}{2}} = \left(X \left| X \right. \right)^{\frac{1}{2}}. \tag{A12}$$

2. Reduced Density Operators

The space-spin integral kernel, $D^p\left(\boldsymbol{\kappa}_p, \boldsymbol{\kappa}'_p\right)$ of the p^{th} order reduced density operator of the N-electron state, D, is defined as

$$D^p\left(\boldsymbol{\kappa}_p; \boldsymbol{\kappa}'_p\right) = \mathrm{Tr}\left\{ \Phi\left(\boldsymbol{\kappa}'_p\right)^\dagger \Phi\left(\boldsymbol{\kappa}_p\right) D \right\}, \tag{A13}$$

where

$$\boldsymbol{\kappa}_p \equiv \boldsymbol{x}_1, \cdots, \boldsymbol{x}_p \tag{A14}$$

$$\boldsymbol{x}_j \equiv \boldsymbol{r}_j, \boldsymbol{\xi}_j; 1 \le j \le p$$

and the field operators are defined in Appendix (C). The multi-electron operators $\Phi\left(\boldsymbol{\kappa}_p\right)^\dagger$ are defined as the product of the one-electrons operators $\Phi\left(\boldsymbol{\kappa}_j\right)^\dagger$ i.e.

$$\Phi\left(\boldsymbol{\kappa}_p\right)^\dagger = \Phi\left(\boldsymbol{x}_1\right)^\dagger \cdots \Phi\left(\boldsymbol{x}_p\right)^\dagger. \tag{A15}$$

The space integral kernels are also given by eq.$(A13)$ but with

$$\boldsymbol{\kappa}_p \equiv \boldsymbol{r}_1, \cdots, \boldsymbol{r}_p. \tag{A16}$$

The p^{th}-Order RDO is then given by

$$D^p = \frac{1}{(p!)^2} \int D^p \left(\boldsymbol{\kappa}_p; \boldsymbol{\kappa}'_p\right) |\boldsymbol{\kappa}_p\rangle \left\langle \boldsymbol{\kappa}'_p \right| d\boldsymbol{\kappa}_p d\boldsymbol{\kappa}'_p, \tag{A17}$$

where $\left\{|\boldsymbol{\kappa}_p\rangle, \left\langle \boldsymbol{\kappa}'_p\right|\right\}$ are to be understood as generalized bases (Appendix (B)).

The RDO kernel for a state, D, that does not have a fixed number of electrons can be given in terms of the kernels for pure number states as

$$D(\boldsymbol{\kappa}; \boldsymbol{\kappa}') = \sum_{0 \le p \le \infty} \alpha_p D^p \left(\boldsymbol{\kappa}_p; \boldsymbol{\kappa}'_p\right); \sum_{1 \le p \le \infty} \alpha_j = 1; 0 \le \alpha_j \le 1 \tag{A18}$$

3. Reduced Density Matrices (RDM)

The matrix elements $D^p_{j_1 \cdots j_p k_1 \cdots k_p}$ of the p^{th}-Order RDM $\mathbb{D}^p$ with respect to the complete orthonormal basis $\{|\varphi_j\rangle| \, 1 \le j_1 \le \cdots \le j_p \le \infty\}$ of $\mathcal{H}^1$ are given by

$$D^p_{j_1 \cdots j_p k_1 \cdots k_p} = \int \varphi_{j_1}(\boldsymbol{x}_1)^* \cdots \varphi_{j_p}(\boldsymbol{x}_p)^* D^p \left(\boldsymbol{\kappa}_p; \boldsymbol{\kappa}'_p\right) \varphi_{k_1}(\boldsymbol{x}'_1) \cdots \varphi_{k_p}\left(\boldsymbol{x}'_p\right) d\boldsymbol{\kappa}_p d\boldsymbol{\kappa}'_p \tag{A19}$$

$$1 \le j_1 \le \cdots \le j_p \le \infty; 1 \le k_1 \le \cdots \le k_p \le \infty. \tag{A20}$$

Thus

$$D^p = \sum_{\substack{1 \le j_1 \le \cdots \le j_p \le \infty \\ 1 \le k_1 \le \cdots \le k_p \le \infty}} D^p_{j_1 \cdots j_p k_1 \cdots k_p} \left|\varphi_{j_1} \cdots \varphi_{j_p}\right\rangle \left\langle \varphi_{k_1} \cdots \varphi_{k_p}\right|. \tag{A21}$$

$\lceil \left|\varphi_{j_1} \cdots \varphi_{j_p}\right\rangle = \frac{1}{\sqrt{p!}} \varphi_{j_1} \wedge \cdots \wedge \varphi_{j_p} = \mathcal{A}_p \varphi_{j_1} \otimes \cdots \otimes \varphi_{j_p}$ thus the matrix elements are given by expectation value with respect to the simple tensor product i.e.

$$\left\langle \varphi_{j_1} \cdots \varphi_{j_p}\right| D^p \varphi_{k_1} \cdots \varphi_{k_p}\rangle = \frac{1}{p!} \left\langle \varphi_{j_1} \wedge \cdots \wedge \varphi_{j_p}\right| D^p \varphi_{k_1} \wedge \cdots \wedge \varphi_{k_p}\rangle \tag{A22}$$

$$= \frac{1}{p!} \left\langle \mathcal{A}_p \varphi_{j_1} \otimes \cdots \otimes \varphi_{j_p}\right| D^p \mathcal{A}_p \varphi_{k_1} \otimes \cdots \otimes \varphi_{k_p}\rangle = \frac{1}{p!} \left\langle \varphi_{j_1} \otimes \cdots \otimes \varphi_{j_p}\right| \mathcal{A}_p D^p \mathcal{A}_p \varphi_{k_1} \otimes \cdots \otimes \varphi_{k_p}\rangle \tag{A23}$$

$$= \frac{1}{p!} \left\langle \varphi_{j_1} \otimes \cdots \otimes \varphi_{j_p}\right| D^p \varphi_{k_1} \otimes \cdots \otimes \varphi_{k_p}\rangle \rfloor \tag{A24}$$

APPENDIX B: GENERALIZED BASES

Generalized bases make use of the "rigged Hilbert space" construction [**?**]. A rigging of a Hilbert space $\mathcal{H}$ is produced by a dense (in the Hilbert space norm) Topological Vector Space (TVS) space $\mathcal{R} \subset \mathcal{H}$, which has a dual $\mathcal{R}^d$ such that $\mathcal{H} = \mathcal{H}^d$ can be embedded in $\mathcal{R}^d$ i.e.

$\mathcal{H} \subset \mathcal{R}^d$ produing a Gelfand Triple $\mathcal{R} \subset \mathcal{H} \subset \mathcal{R}^d$. Many topologies may be put on $\mathcal{R}$ (that are finer than the Hilbert space topology) producing different $\mathcal{R}^d$'s, the only restriction is that the embedding $i : \mathcal{H} \rightarrow \mathcal{R}^d$ is antilinear, injective and continuous. Different $\mathcal{R}$'s produce different riggings. A space that is often chosen for $\mathcal{R}$ is $\mathcal{S}(\mathbb{R}^n)$ the space of rapidly decreasing test functions on $\mathbb{R}^n$, its dual space $\mathcal{S}(\mathbb{R}^n)^d$ is the space of tempered distributions, which contains Dirac delta functions, $\delta_{\mathbf{r}}$, and plane waves $e^{i\mathbf{k}\cdot\mathbf{r}}$ [**?**].

As $\mathcal{R}^d$ is the dual of $\mathcal{R}$ one denotes the elements of $\mathcal{R}^d$ as $\langle a'|$ and those of $\mathcal{R}$ as $|a\rangle$ .As $\mathcal{R}$ and $\mathcal{H}$ can be embedded in $\mathcal{R}^d$ one can always consider $\langle a| \,; a \in \mathcal{R}$ and $\langle h| \,; h \in \mathcal{H}$ with the understanding that $\langle a|$ are defined on $\mathcal{R}$ and $\langle h|$ on $\mathcal{H}$.

The Nuclear Spectral Theorem [**?**] states that any self-adjoint operator A (in general not bounded) has an expansion

$$A = \int_{\sigma(A)} a\,|a\rangle\,\langle a|\,da, \tag{B1}$$

$\lceil$where the measure da can be expanded as

$$da = da_{ac} + da_{pp} + da_{sc} = da_{ac} + da_{sc} + \sum_{1\leq j\leq r} \delta\left(\alpha_j - a\right) da. \tag{B2}$$

One should observe that if there are sets/points in the spectrum that have Lesbague measure (da_{ac}) zero then they have non zero da_{sc} measure.$\rfloor$

in the sense that

$$\langle\varphi\,|A\psi\rangle = \int_{\sigma(A)} a\,\langle\varphi|\,a\rangle\,\langle a|\,\psi\rangle\,da \tag{B3}$$

$$\langle\varphi|\,a\rangle = \overline{\langle a|\,\varphi\rangle}, \tag{B4}$$

where $\{|\varphi\rangle\,,\langle\psi|\} \in \mathcal{R}$, $\{|a\rangle\,,\langle a|\} \in \mathcal{R}^d$, the spectrum of $\sigma(A) \subseteq \mathbb{R}$ and da is a point measure. All of the vectors $|a\rangle$ are eigenvectors of A i.e.

$$A\,|a\rangle = a\,|a\rangle\,, \tag{B5}$$

again in the sense

$$\langle\varphi\,|Aa\,\rangle = a\,\langle\varphi\,|a\,\rangle = a\overline{\langle\varphi\,|a\,\rangle}, |\varphi\rangle \in \mathcal{R}. \tag{B6}$$

The set of vectors $\{\,|a\rangle|\,a \in \sigma(A)\}$ forms a generalized basis of $\mathcal{H}$ and any vector $|v\rangle \in \mathcal{R}$ has the expansion

$$|v\rangle = \int_{\sigma(A)} \langle a|\,v\rangle\,|a\rangle\,da; |a\rangle \in \mathcal{R}^d \tag{B7}$$

The basis $\{ |a\rangle | \, a \in \sigma(A)\}$ is orthonormal in the sense

$$\langle a_1 | \, \nu\rangle = \int_{\sigma(A)} \langle a_1 | \, a_2\rangle \langle a_2 | \, v\rangle \, da_2 \forall |v\rangle \in \mathcal{R} \tag{B8}$$

so that $\langle a_1 | \, a_2\rangle$ can be identified as

$$\langle a_1 | \, a_2\rangle = \delta_{a_1 - a_2}. \tag{B9}$$

The vectors $|v\rangle \in \mathcal{R}$ have unique expansion coefficients $\langle a | \, v\rangle$ (if the spectral measure has no continuously singular parts) i.e.

$$|v_1\rangle = |v_2\rangle \Leftrightarrow \langle a | \, v_1\rangle = \langle a | \, v_2\rangle \, \forall a \tag{B10}$$

$$\left\langle a' \middle| \int_{\sigma(A)} \langle a | \, v_1\rangle \, |a\rangle \, da \right\rangle = \left\langle a' \middle| \int_{\sigma(A)} \langle a | \, v_2\rangle \, |a\rangle \, da \right\rangle \tag{B11}$$

$$\int_{\sigma(A)} \langle a | \, v_1\rangle \, \delta(a' - a) \, da = \int_{\sigma(A)} \langle a | \, v_2\rangle \, \delta(a' - a) \, da \tag{B12}$$

$$\langle a' | \, v_1\rangle = \langle a' | \, v_2\rangle \, . \tag{B13}$$

Thus

$$|v_1\rangle - |v_2\rangle = 0 \Rightarrow \int_{\sigma(A)} \{\langle a | \, v_1\rangle - \langle a | \, v_2\rangle\} \, \delta(a' - a) \, da = 0 \tag{B14}$$

in contrast to

$$\int_{\sigma(A)} |v_1(a) - v_2(a)|^2 \, da = 0 \Leftrightarrow v_1(a) \overset{a.e}{=} v_2(a) \text{ wrt } da \tag{B15}$$

as $|v_1\rangle$ and $|v_2\rangle$ do not have to be in $\mathcal{R}$.

If the operator A is a compact and self-adjoint (thus bounded) then the basis $\{ |a\rangle | \, a \in \sigma(A);\}$ is a conventional orthonormal basis contained in $\mathcal{H}$ i.e. $|a\rangle \in \mathcal{H} \, \forall a \in \sigma(A)$.

The operators $|a\rangle \langle a'|$ can thus be interpretted as

$$|a\rangle \langle a'| : \mathcal{R}^d \to \mathcal{R}^d. \tag{B16}$$

In particular consider the position operator $\mathbf{Q}$

$$\mathbf{Q} = \int_{\mathbb{R}^3} \mathbf{r} \, |\mathbf{r}\rangle \langle \mathbf{r}| \, d\mathbf{r}, \tag{B17}$$

then any state, $|\varphi\rangle$, that has a wavefunction, $\langle \mathbf{r} | \, \varphi\rangle$, belonging to the test function space $\mathcal{R}$ $= \mathcal{S}(\mathbb{R}^n)$, can be expanded as

$$|\varphi\rangle = \int_{\mathbb{R}^3} \langle \mathbf{r} | \, \varphi\rangle \, |\mathbf{r}\rangle \, d\mathbf{r}, \tag{B18}$$

where

$$\langle \mathbf{r} | \, \mathbf{r}' \rangle = \delta_{\mathbf{r}-\mathbf{r}'}. \tag{B19}$$

(Note that $\mathcal{R} \neq \mathcal{H}$ as there are functions in $\mathcal{H}$ that are not bounded on compact sets and hence $\langle \mathbf{r} | \, \varphi \rangle$ is not defined for all $|\varphi\rangle \in \mathcal{H}$).

⌈

$$|\varphi\rangle = |\psi\rangle \quad \Leftrightarrow \{ \langle \varphi | \, \chi \rangle = \langle \psi | \, \chi \rangle \equiv \langle \varphi | \, \chi \rangle - \langle \psi | \, \chi \rangle = 0 \} \, \forall \chi \in \mathcal{R} \tag{B20}$$

⌋

APPENDIX C: SECOND QUANTIZED DENSITY OPERATORS

Fermion second quantized operators $\left\{ a_j^\dagger \middle| \, 1 \leq j \leq \infty \right\}$ are defined by the action on the vacuum state $|\phi\rangle$ by

$$a_j^\dagger \, |\phi\rangle = |\varphi_j\rangle \, , 1 \leq j \leq \infty, \tag{C1}$$

where $\{ |\varphi_j\rangle | \, 1 \leq j \leq \infty \}$ is a complete orthonormal basis of one-electron space $\mathcal{H}^1$ and $\left\{ a_j^\dagger, a_k \middle| \, 1 \leq j, k \leq \infty \right\}$ satisfy the canonical anti-commutation relationships

$$\left[a_j^\dagger, a_k \right]_+ = a_j^\dagger a_k + a_k a_j^\dagger = \delta_{jk}; 1 \leq j, k \leq \infty. \tag{C2}$$

The operators $\left\{ a_j, a_j^\dagger \middle| \, 1 \leq j \leq \infty \right\}$ can be considered as the values of the maps

$$\Phi, \Phi^\dagger : \mathcal{H}^1 \to \mathcal{B} \left(\mathcal{H}_{\mathcal{F}} \right) \tag{C3}$$

given by

$$a_j = \Phi \left(\varphi_j \right), \tag{C4}$$

$$a_j^\dagger = \Phi \left(\varphi_j \right)^\dagger, \tag{C5}$$

where the Fock space, $\mathcal{H}_{\mathcal{F}}$, is defined by

$$\mathcal{H}_{\mathcal{F}} = \bigoplus_{0 \leq N \leq \infty} \wedge^N \mathcal{H}^1. \tag{C6}$$

The point field operators $\{ \Phi(\boldsymbol{r}) | \, \boldsymbol{r} \in \mathbb{R}^3 \}$ are defined by [**?**]

$$\Phi(\boldsymbol{r}) = \sum_{1 \leq j \leq \infty} \varphi_j(\boldsymbol{r})^* \, a_j \tag{C7}$$

and the density operators, $\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r})$, formally by

$$\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r}) = \sum_{1\le j,k\le\infty} \varphi_j(\boldsymbol{r})\,\varphi_k(\boldsymbol{r})^{*}\,a_j^{\dagger}a_k. \tag{C8}$$

The unbounded operators $\Phi(\boldsymbol{r})$ are densely defined (but not closeable) on the Fock space, $\mathcal{H}_{\mathcal{F}}$, nor the subspaces $\mathcal{H}^N = \wedge^N\mathcal{H}^1$, while the domains of their adjoints $\Phi(\boldsymbol{r})^{\dagger}$ are empty i.e. do not contain any vectors belonging to $\mathcal{H}_{\mathcal{F}}$ [**?**]. However, matrix elements, $\mathrm{Tr}\left\{A^{\dagger}\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r})B\right\}$, of the density super operators $\widehat{\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r})}$ can be defined as the distributional limits of $\mathrm{Tr}\left\{A^{\dagger}\Phi(f)^{\dagger}\Phi(f)B\right\}$ as $f\to\delta_{\boldsymbol{r}}$ where

$$\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r}) \equiv \Phi(\delta_{\boldsymbol{r}})^{\dagger}\Phi(\delta_{\boldsymbol{r}}). \tag{C9}$$

Even though $\Phi(\boldsymbol{r})$ are not closeable the operators $\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r})$ are bounded as

$$\left\|\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r})\right\| = \left[\sup_{\substack{\Psi\in\mathcal{H}_{\mathcal{F}}\\ \|\Psi\|=1}}\left\{\left\langle\Psi\left|\Phi(\boldsymbol{r})^{\dagger}\Phi(\boldsymbol{r})\Psi\right.\right\rangle\right\}\right]^{\frac{1}{2}} = \sup_{\substack{\Psi\in\mathcal{H}_{\mathcal{F}}\\ \|\Psi\|=1}}\{\rho_{\Psi}(\boldsymbol{r})\}^{\frac{1}{2}} < \infty \tag{C10}$$

as all states $|\Psi\rangle\langle\Psi| \in \mathcal{B}_{1sa+}(\mathcal{H}_{\mathcal{F}})$ thus $\rho_{\Psi}(\boldsymbol{r}) \le \int \rho_{\Psi}(\boldsymbol{r})\,d\boldsymbol{r} < \infty$.?

APPENDIX D: SUPER OPERATORS

Super-operators (which could be unbounded), $\widehat{T}$, acting on the Hilbert space $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$, i.e.

$$\widehat{T} : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to \mathcal{B}_{2sa}\left(\mathcal{H}^N\right), \tag{D1}$$

can be generated by self adjoint operators belonging to $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$

$$T : \mathcal{H}^N \to \mathcal{H}^N \tag{D2}$$

by

$$\widehat{T}(Y) = \frac{1}{2}[T,Y]_{+} \equiv \frac{1}{2}(TY+YT). \tag{D3}$$

Noting that

$$\mathrm{Tr}\{TYX\} = \mathrm{Tr}\{XTY\} = \mathrm{Tr}\{YTX\} = \mathrm{Tr}\{XYT\} \text{ as } X,Y,T \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \tag{D4}$$

one can observe that

$$\frac{1}{2}(\mathrm{Tr}\{TYX\} + \mathrm{Tr}\{YTX\}) = \mathrm{Tr}\{TYX\} \tag{D5}$$

$$\frac{1}{2}(\mathrm{Tr}\{XTY\} + \mathrm{Tr}\{XYT\}) = \mathrm{Tr}\{TYX\} \tag{D6}$$

leading to

$$\left(\widehat{T}(Y)\Big| X\right) = (TY|X). \tag{D7}$$

The operator $\widehat{T}$ is self adjoint as

$$\left(\widehat{T}X|Y\right) = \frac{1}{2}\operatorname{Tr}\left\{[T,X]_+ Y\right\} = \frac{1}{2}\operatorname{Tr}\{TXY + XTY\} \tag{D8}$$

$$= \frac{1}{2}\operatorname{Tr}\{YTX + TYX\} = \frac{1}{2}\operatorname{Tr}\{XTY + XYT\} \tag{D9}$$

$$= \left(X\Big|\widehat{T}Y\right). \tag{D10}$$

The domain, $\operatorname{Dom}\left\{\widehat{T}\right\}$, of $\widehat{T}$

$$\operatorname{Dom}\left\{\widehat{T}\right\} = \left\{Y\,\middle|\operatorname{Tr}\left\{\left(\widehat{T}(Y)\right)^2\right\} < \infty\right\} \equiv \frac{1}{4}\{Y|\operatorname{Tr}\{(TY+YT)(TY+YT)\} < \infty\} \tag{D11}$$

can be generated by projectors, $|\Phi\rangle\langle\Phi|$ onto normalized states $|\Phi\rangle$ that lie in the domain, $\operatorname{Dom}\{T\}$, of T.This domain is dense in $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ if $\operatorname{Dom}\{T\}$ is dense in $\mathcal{H}^N$ as

$$\left\{|\Phi\rangle\langle\Phi|\,\middle|\operatorname{Tr}\left\{\left(\widehat{T}(|\Phi\rangle\langle\Phi|)\right)^2\right\} = \frac{1}{2}\left\{\langle\Phi|T\Phi\rangle^2 + \left\langle\Phi|T^2\Phi\right\rangle\right\} = \|T\|^2 < \infty\ ; |\Phi\rangle \in \operatorname{Dom}\{T\}\right\}$$

and the domain of T is generated by the set of HS convergent real linear combinations

$$Y = \sum_{1\leq j\leq\infty} Y_j\,|\Phi_j\rangle\langle\Phi_j| \tag{D12}$$

The operator $\widehat{T}$ could also be defined in terms of the quadratic form

$$\left(X\Big|\widehat{T}Y\right) = \frac{1}{2}\operatorname{Tr}\left\{X[T,Y]_+\right\} = \frac{1}{2}\operatorname{Tr}\{XTY + XYT\} = \frac{1}{2}\operatorname{Tr}\{XTY + YTX\} \tag{D13}$$

$$= \frac{1}{2}\operatorname{Tr}\{XTY + XTY\} = \operatorname{Tr}\{XTY\}\,\forall X,\,Y \in \operatorname{Dom}\left\{\widehat{T}\right\}. \tag{D14}$$

APPENDIX E: REGULAR POINTS

1. Derivatives

A Banach space mapping, F,between spaces $\mathcal{B}_1$ and $\mathcal{B}_2$ i.e.

$$F : \mathcal{B}_1 \to \mathcal{B}_2, \tag{E1}$$

is Gâteaux differentiable at a point $A \in \mathcal{B}_1$ contained in an open subset U of $\mathcal{B}_1$ if

$$\nabla_B F(A) = \lim_{t \to 0} \frac{1}{t} \{F(A + tB) - F(A)\} = \left. \frac{d}{dt} F(A + tB) \right|_{t=0}, \tag{E2}$$

exists for any $B \in \mathcal{B}_1$. If $\nabla F(A)$ exists for all $B \in \mathcal{B}_1$ it is called the Gâteaux derivative of F at A and $\nabla_B F(A) = [\nabla F(A)](B)$ the directional Gâteaux derivative in the direction B at the point A. The map F is called Fréchet differentiable at a point $A \in \mathcal{B}_1$ if there exists a linear operator

$$dF(A) : \mathcal{B}_1 \to \mathcal{B}_2 \tag{E3}$$

such that

$$\lim_{\|B\| \to 0} \frac{1}{\|B\|} \{F(A + B) - F(A) - [dF(A)](B)\} = 0, \tag{E4}$$

$dF(A)$ is called the Fréchet derivative of F at A. F is Fréchet differentiable in an open subset U of $\mathcal{B}_1$if and only if it is

- Gâteaux differentiable i.e. $\nabla F(A)$ exists
- $[\nabla F(A)](B) \equiv \nabla_B F(A)$ is linear in B
- $\sup_{B \neq 0} \left\{ \frac{\|\nabla_B F(A)\|}{\|B\|} \right\} < \infty$

then

$$\nabla_B F(A) = [dF(A)](B) = [\nabla F(A)](B). \tag{E5}$$

Fréchet differentiation defines a map (generally non linear)

$$d : \mathcal{F}(\mathcal{B}_1, \mathcal{B}_2) \to \mathcal{F}(\mathcal{B}_1, \mathcal{B}_2) \tag{E6}$$

$$d : F \to dF. \tag{E7}$$

that produces the Fréchet derivative, the Fréchet derivative at a point A is $dF(A) \in \mathcal{L}(\mathcal{B}_1, \mathcal{B}_2)$, the bounded linear maps from $\mathcal{B}_1$ to $\mathcal{B}_2$.

2. Tangent and Normal Spaces

The constraint functional is a quadratic map

$$\Xi_2 : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to \mathfrak{D} \subset L_{1+}\left(\mathbb{R}^3\right) \cap L_{3+}\left(\mathbb{R}^3\right) \subset L_1\left(\mathbb{R}^3\right), \tag{E8}$$

where

$$\Xi_2(A) = \left(A \middle| \widehat{\Phi^\dagger\Phi}_\delta A\right) = \rho. \tag{E9}$$

The super operator valued distribution $\widehat{\Phi^\dagger\Phi}_\delta$ is defined by

$$\widehat{\Phi^\dagger\Phi}_\delta : \mathbb{R}^3 \cong \mathcal{D}_\delta \subset \mathcal{S}\left(\mathbb{R}^3\right)^d \to \mathcal{B}_{sa}\left(\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)\right) \tag{E10}$$

$$\widehat{\Phi^\dagger\Phi}_\delta(\boldsymbol{r}) \equiv \widehat{\Phi^\dagger\Phi}(\delta_{\mathbf{r}}) \in \mathcal{B}_{sa}\left(\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)\right), \tag{E11}$$

(where we have used the identification of $\mathbb{R}^3$ with the set of Dirac delta functions $\mathcal{D}_\delta$), in terms of the operator valued distribution $\Phi^\dagger\Phi_\delta$

$$\Phi^\dagger\Phi_\delta : \mathbb{R}^3 \to \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \tag{E12}$$

$$\Phi^\dagger\Phi_\delta(\boldsymbol{r}) \equiv \Phi^\dagger(\delta_{\mathbf{r}})\,\Phi(\delta_{\mathbf{r}}) \equiv \Phi^\dagger(\boldsymbol{r})\,\Phi(\boldsymbol{r}) \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \tag{E13}$$

as

$$\widehat{\Phi^\dagger\Phi}_\delta(\boldsymbol{r})\,|A) = \left|\left[\Phi(\boldsymbol{r})^\dagger\,\Phi(\boldsymbol{r})\,A\right]_+\right). \tag{E14}$$

The density ρ_A associated with an operator $A \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ is given by

$$\left(\left[\Phi(\boldsymbol{r})^\dagger\,\Phi(\boldsymbol{r})\,A\right]_+ \middle| A\right) = \mathrm{Tr}\left\{\Phi(\boldsymbol{r})^\dagger\,\Phi(\boldsymbol{r})\,A^2\right\}\ \forall\,\boldsymbol{r} \in \mathbb{R}^3. \tag{E15}$$

One does not distinguish between densities that are only different on sets of measure zero i.e.

$$\rho \sim \rho' \text{ if } \int |\rho(\boldsymbol{r}) - \rho'(\boldsymbol{r})|\,d\boldsymbol{r} = 0 \text{ i.e. } \rho \overset{a.e}{=} \rho'. \tag{E16}$$

It is easy to see that Ξ_2 is Fréchet differentiable as the directional Gâteaux derivative, $\nabla_B \Xi_2(A)$, of Ξ_2 exists at A for all B

$$\lim_{t\to 0}\frac{1}{t}\left\{\Xi_2(A+tB) - \Xi_2(A)\right\} = \lim_{t\to 0}\frac{1}{t}\left\{\left((A+tB)\middle|\widehat{\Phi^\dagger\Phi}_\delta(A+tB)\right) - \left(A\middle|\widehat{\Phi^\dagger\Phi}_\delta A\right)\right\} \tag{E17}$$

$$= \lim_{t\to 0}\left\{2\left(A\middle|\widehat{\Phi^\dagger\Phi}_\delta B\right) - t\left(B\middle|\widehat{\Phi^\dagger\Phi}_\delta B\right)\right\} = 2\left(A\middle|\widehat{\Phi^\dagger\Phi}_\delta B\right) \tag{E18}$$

leading to a Gâteaux derivative

$$\nabla\Xi_2(A) = 2\left(\left[\Phi^\dagger\Phi_\delta(A)\right]_+\right|, \tag{E19}$$

which is clearly a bounded linear functional

$$\nabla\Xi_2(A) : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to L_1\left(\mathbb{R}^3\right). \tag{E20}$$

The three conditions for Fréchet differentialability are satisfied and the Fréchet derivative of Ξ_2 at A is then also

$$d\Xi_2(A) = \nabla\Xi_2(A) = 2\left(\left[\Phi^\dagger\Phi_\delta(A)\right]_+\right| \tag{E21}$$

and

$$d\Xi_2(A) : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to L_1\left(\mathbb{R}^3\right). \tag{E22}$$

The differential map

$$d\Xi_2 : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to \mathcal{L}\left(\mathcal{B}_{2sa}\left(\mathcal{H}^N\right), L_1\left(\mathbb{R}^3\right)\right) \tag{E23}$$

is linear (as Ξ_2 is quadratic) and one can consider

$$\operatorname{Ker} d\Xi_2 = \{\, A|\, d\Xi_2(A) = 0\} \tag{E24}$$

and

$$\operatorname{Ran} d\Xi_2 = \{\, B|\, B = d\Xi_2(A)\}. \tag{E25}$$

$A \in \operatorname{Ker} d\Xi_2$ if

$$\left(\left[\Phi^\dagger(\boldsymbol{r})\,\Phi(\boldsymbol{r})\,A\right]_+\right| = 0\,\forall\,\boldsymbol{r}\in\mathbb{R}^3 \Longleftrightarrow A = 0, \tag{E26}$$

so is empty. However $\operatorname{Ker} d\Xi_2(A)$ is not empty as

$$\left(A\middle|\Phi^\dagger\widehat{(\boldsymbol{r})\,\Phi}(\boldsymbol{r})B\right) = \left(A\middle|\left[\Phi^\dagger(\boldsymbol{r})\,\Phi(\boldsymbol{r})\,B\right]_+\right) = \operatorname{Tr}\left\{A\left[\Phi(\boldsymbol{r})^\dagger\,\Phi(\boldsymbol{r})\,B\right]_+\right\} \tag{E27}$$

$$= \operatorname{Tr}\left\{A\Phi(\boldsymbol{r})^\dagger\,\Phi(\boldsymbol{r})\,B\right\} = 0 \quad \forall\boldsymbol{r}\in\mathbb{R}^3 \tag{E28}$$

implies that A is localized on $\boldsymbol{\Delta}_A$ and B on $\boldsymbol{\Delta}_B$ and

$$\boldsymbol{\Delta}_A \cap \boldsymbol{\Delta}_B = \phi, \tag{E29}$$

where

$$\boldsymbol{\Delta}_X = \operatorname{supp}\rho_X = \overline{\{\mathbf{r}|\,\rho_X(\mathbf{r}) \neq 0\}}, \tag{E30}$$

i.e. the closure of the set of points where ρ_X is nonzero. $\boldsymbol{\Delta}_X$ might or might not be a compact i.e. a closed and bounded set.

The operators $\left\{\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)\right|\mathbf{r}\in\mathbb{R}\right\}$ form a generalized basis for ? $\mathcal{B}_{2sa}\left(L_{2}\left(\boldsymbol{\Delta}_{A}\right)^{N}\right)$, which can be seen by considering the projectors

$$\mathcal{P}=\int_{\boldsymbol{\Delta}_{A}}\frac{\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right|}{\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\middle|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)}d\boldsymbol{r} \tag{E31}$$

$$=\int_{\boldsymbol{\Delta}_{A}}\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right|\frac{1}{\rho_{A^{2}}\left(\boldsymbol{r}\right)}d\boldsymbol{r}, \tag{E32}$$

which have the effect

$$|B)=\int_{\boldsymbol{\Delta}_{A}}\frac{\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\middle|B\right)}{\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\middle|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)}\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)d\boldsymbol{r} \tag{E33}$$

$$=\int_{\boldsymbol{\Delta}_{A}}\frac{\mathrm{Tr}\left\{\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}B\right\}}{\mathrm{Tr}\left\{\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right\}}\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)d\boldsymbol{r} \tag{E34}$$

$$=\int_{\boldsymbol{\Delta}_{A}}\mathrm{Tr}\left\{A\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)B\right\}\left|\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right)\frac{1}{\rho_{A^{2}}\left(\boldsymbol{r}\right)}d\boldsymbol{r} \tag{E35}$$

if $\boldsymbol{\Delta}_{B}\subseteq\boldsymbol{\Delta}_{A}$. If $\boldsymbol{\Delta}_{A}\cap\boldsymbol{\Delta}_{B}=\phi$ then $A\perp B$ i.e.

$$\left(A\right|B)=0. \tag{E36}$$

The map Ξ_{2} defines a manifold in $\mathcal{B}_{2sa}\left(\mathcal{H}^{N}\right)$ given by

$$\mathfrak{M}=\left\{A\right|\Xi_{2}\left(A\right)=\rho\}. \tag{E37}$$

The tangent space $\mathcal{T}_{A}$ at a point A of $\mathfrak{M}$ is given by

$$\mathcal{T}_{A}=\left\{B\left|\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\middle|B\right)=0\forall\ \boldsymbol{r}\in\mathbb{R}^{3}\right.\right\}\subseteq\mathrm{Ker}\,d\Xi_{2}\left(A\right) \tag{E38}$$

and the normal space $\mathcal{N}_{A}$ by

$$\mathcal{N}_{A}=\left\{B\left|\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\middle|B\right)\neq0\forall\ \boldsymbol{r}\in\mathbb{R}^{3}\right.\right\}. \tag{E39}$$

One can show that $\left\{\left[d\Xi_{2}\left[\boldsymbol{r}\right]\left(A\right)\right]\left(\cdot\right)\equiv\left(\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\right|\equiv\mathrm{Tr}\left\{\left[\Phi\left(\boldsymbol{r}\right)^{\dagger}\Phi\left(\boldsymbol{r}\right)A\right]_{+}\cdot\right\}\middle|\ \boldsymbol{r}\in\mathbb{R}^{3}\right\}$ is a generalized basis for operators $\in\mathrm{Ker}\,d\Xi_{2}\left(A\right)^{\perp}$, (see Appendix basis B), for $\mathcal{N}_{A}$ which shows that

$$\mathcal{B}_{2sa}\left(\mathcal{H}^{N}\right)=\mathcal{T}_{A}\oplus\mathcal{N}_{A} \tag{E40}$$

and

$$[d\Xi_2(A)] : \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \to L_1(\mathbf{\Delta}_A) \tag{E41}$$

is surjective so that A is a *regular* [**?**] point of the constraint.

APPENDIX F: COMPLEXIFICATION OF $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$

The complex Hilbert space $\mathcal{B}_2\left(\mathcal{H}^N\right)$ can be considered as the complexification of the real Hilbert space $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ i.e.

$$\mathcal{B}_2\left(\mathcal{H}^N\right) = \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \oplus i\mathcal{B}_{2sa}\left(\mathcal{H}^N\right), \tag{F1}$$

such that $Z \in \mathcal{B}_2\left(\mathcal{H}^N\right)$, where

$$Z = X + iY;\ X, Y \in \mathcal{B}_{2sa}\left(\mathcal{H}^N\right) \tag{F2}$$

The inner product in $\mathcal{B}_2\left(\mathcal{H}^N\right)$ is defined in terms of the inner product in $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ as

$$(Z_1 | Z_2)_{HS} = (X_1 | X_2) + (Y_1 | Y_2) + i\left\{(X_1 | Y_2) - (Y_1 | X_2)\right\}, \tag{F3}$$

which corresponds to

$$(Z_1 | Z_2)_{HS} = \mathrm{Tr}\left\{Z_1^\dagger Z_2\right\}, \tag{F4}$$

as

$$\mathrm{Tr}\left\{(X_1 - iY_1)(X_2 + iY_2)\right\} = \mathrm{Tr}\left\{X_1X_2 - iY_1X_2 + iX_1Y_2 + Y_1Y_2\right\}. \tag{F5}$$

Note that $\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ (self adjoint operators) and $i\mathcal{B}_{2sa}\left(\mathcal{H}^N\right)$ (anti self adjoint operators) are not orthogonal subspaces (they are not even subspaces under $\mathbb{C}$-linear combination).

APPENDIX G: STATE NORMALIZATION

If $D^N \in \mathcal{B}_2\left(\mathcal{H}^N\right)$ then

$$\mathrm{Tr}\left\{\mathcal{N} D^N\right\} = N\,\mathrm{Tr}\left\{D^N\right\}, \tag{G1}$$

where

$$\mathcal{N} = \int \Phi(\boldsymbol{r})^\dagger\, \Phi(\boldsymbol{r})\, d\boldsymbol{r}, \tag{G2}$$

as

$$\mathcal{N}\,|\Psi\rangle = N\,|\Psi\rangle \quad \forall\,|\Psi\rangle \in \mathcal{H}^N. \tag{G3}$$

For any state D

$$\operatorname{Tr}\{\mathcal{N}D\} = \int \rho\left(\boldsymbol{r},\boldsymbol{\xi}\right) d\boldsymbol{r} d\boldsymbol{\xi}, \tag{G4}$$

thus

$$N\operatorname{Tr}\left\{D^N\right\} = N \Rightarrow \operatorname{Tr}\left\{D^N\right\} = 1. \tag{G5}$$

APPENDIX H: DOES $\rho_1 = \rho_2$ POINTWISE OR IN THE MEAN

When do we consider two densities to be equal i.e.

$$\rho_1 = \rho_2 \tag{H1}$$

is it when

$$\rho_1\left(\mathbf{r}\right) = \rho_2\left(\mathbf{r}\right) \;\; \forall\mathbf{r} \in\mathbb{R}^3 \tag{H2}$$

or

$$\int \left|\rho_1\left(\mathbf{r}\right) - \rho_2\left(\mathbf{r}\right)\right| d\mathbf{r} = 0 \text{ i.e. } \rho_1 \overset{a.e}{=} \rho_2? \tag{H3}$$

Two FORDO's are equal when

$$\left(A\left|D^1\right.\right) = \left(A\left|D^2\right.\right) \;\; \forall\mathbf{A} \in\mathcal{B}_{sa}\left(\mathcal{H}^1\right) \Longleftrightarrow \tag{H4}$$

$$\left(A\left|D_1^1 - D_2^1\right.\right) = 0 \;\forall\mathbf{A} \in\mathcal{B}_{sa}\left(\mathcal{H}^1\right);\; D_1^1 - D_2^1 \in \mathcal{B}_{1sa}\left(\mathcal{H}^1\right), \tag{H5}$$

thus $D_1^1 - D_2^1 = 0$ as an operator in $\mathcal{B}_{1sa}\left(\mathcal{H}^1\right)$ i.e.

$$\int \left|D_1^1\left(\mathbf{r}_1,\mathbf{r}_2\right) - D_2^1\left(\mathbf{r}_1,\mathbf{r}_2\right)\right| d\mathbf{r}_1 d\mathbf{r}_2 = 0 \equiv D_1^1\left(\mathbf{r}_1,\mathbf{r}_2\right) \overset{a.e}{=} D_2^1\left(\mathbf{r}_1,\mathbf{r}_2\right) \tag{H6}$$

so in particular

$$\int \left\{\rho_1\left(\mathbf{r}\right) - \rho_2\left(\mathbf{r}\right)\right\} d\mathbf{r} = 0 \equiv \rho_1 \overset{a.e}{=} \rho_2. \tag{H7}$$

APPENDIX I: SPIN

APPENDIX J: SYMMETRIC FUNCTION RELATIONSHIPS

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